



WSL&WDL

De-energized Tap-changer

Operating Instructions

HM 0.460.602-03.07/2022

IMPORTANT:
*Please read this manual carefully
before using the tap-changer and
save it for reference.*



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Before using, please read this instruction carefully. The product may have been modified since this document was published. We reserve the right to change the technical data, design and scope of supply without warning in advance. Generally, information provided with individual quotations and orders is binding. Huaming has no liability for any damage due to incorrect interpretation of this document.

The original document is written in Chinese.

Keep this manual at hand for future use and in case of any question, please contact Huaming.

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The original operating instructions were drawn up in Chinese.

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1.Foreword

This instruction manual is a comprehensive description of product transportation, installation, commissioning, test and maintenance. It is intended for trained and qualified staff related to transformer production, testing and quality assurance, and has reference value for transformer design and operation personnel.

1.1 Manufacturer

The product is manufactured by:

Shanghai Huaming Power Equipment Co., Ltd.

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Further information on the product and copies of this technical file are available from this address if required.

1.2.Subject to change without notice

The information contained in this technical file comprise the technical specifications released at the time of printing. Significant modifications will be included in a new edition of the technical file. The document and version numbers for this technical file are shown in the footer

1.3 Safety

Safety instructions are explained in this manual. They must be observed and followed to avoid relevant dangers.

Personal safety equipment must be worn during work to prevent potential risks.

	Protective clothing to protect against hazards in the work environment. Do not wear rings, necklaces or other jewelry.
	Safety shoes to protect against falling sharp & heavy objects and slippery surfaces.
	Safety glasses to protect eyes from flying parts and splashing liquids.
	Safety helmet to protect from falling and flying parts and materials.
	Hearing protection to protect from hearing damage.
	Protective gloves to protect hands from mechanical, chemical, and electrical hazards.

Table 1 Personal safety equipment list

1.4 Hazard communication system

Warnings in this technical file use the following format:



SIGNAL WORD!

Consequences Action

The following signal words are are used:

Signal words	Hazard level	Consequences of failure to comply
Danger	Immediate threat of danger	Death or serious injury certainly occur Warning
Warning	Prebable threat of danger	Death or serious injury apparently occur Warning
Prohibition	Prebable dangerous situation	Minor or moderate injury likely occur
Note	Possible dangerous situation	Material damage may occur

Table 2 Signal words in safety instructions

Pictograms warn of dangers:

Pictogram	Meaning
	Danger The yellow triangle warning sign means life and health may be at risk if neglecting the instruction. Neglecting this warning can lead to serious or fatal injuries.
	Dangerous electrical voltage
	Fire hazard
	General prohibition sign! The red symbol indicates some actions are prohibited.

Table 3 Symbols used in warning notices

1.5 Information system

Information is designed to simplify and improve understanding of particular procedures. In this technical file they are laid out as follows:



NOTE!

The blue note symbol indicates important information that must be noted.

1.6 Safety precautions

Precaution	Action
Electric shock	Always unplug a device before inspection. Stay away from high-voltage equipment and wear protective clothing.
Transformer oil	Collect used transformer oil in dedicated drums. Keep it away from flames or sparks. Do not touch it which may cause damages to health.
Fire	In case of fire, extinguish it with powder, foam or carbon dioxide. Do not use water.
Waste	Wastes are dangerous to the environment or human health. Clean them up in time and use different methods according to the wastes' categories.

Table 4 Safety precaution

1.7 Appropriate use

The product and associated equipment and special tools supplied with it comply with the relevant legislation, regulations and standards, particularly health and safety requirements, applicable at the time of delivery.

If used as intended in compliance with the specified requirements and conditions in this technical file as well as the warning notices in this technical file and attached to the product, then the product does not present any hazards for persons, property or the environment. This applies during the entire lifespan, from delivery through installation and operation to disassembly and disposal.

The operational quality assurance system ensures a consistently high quality standard, particularly when it comes to observance of the health and safety requirements.

Use is considered to be appropriate if

- the product is operated according to this technical file, the agreed delivery conditions and technical data, and
- the associated equipment and special tools supplied with it are used solely for the intended purpose and in accordance with the specifications of this technical file.

- the product is used only with the transformer specified in the order.
- the serial number of the manual drive matches that of the WSL type non-excitation tap switch

1.8 Inappropriate use

Use is considered to be inappropriate if the product is used other than as described in this Chapter 1.7. Shanghai Huaming Power Equipment Co., Ltd does not accept liability for damage from unauthorized or inappropriate changes to the product. Unauthorized changes to the product without consultation with Huaming can lead to personal injury, material damage and operational faults.

1.9 Personnel qualification

The product is designed solely for use in electrical or energy systems and facilities operated by appropriately trained staff. This staff comprises people who are familiar with the installation, assembly, commissioning and operation of such products.

1.10 Operator duty of care

To prevent accidents, faults and damage as well as unacceptable adverse effects on the environment, those responsible for transport, installation, operation, maintenance and disposal of the product or parts of the product must ensure the following:

- All warning and hazard notices are complied with.
- Personnel are instructed regularly in all relevant aspects of operational safety, the Operating Instructions and particularly the safety instructions contained therein.
- Regulations and operating instructions for safe working as well as the relevant instructions for staff procedures in the case of accidents and fires are kept on hand at all times and displayed in the workplace where applicable.
- The product is only used in a sound operational condition and safety equipment in particular is checked regularly for operational reliability.
- Use only replacement parts, lubricants and aids which are authorized by the manufacturer.
- Comply with the specified operating conditions and requirements of the installation location.
- All necessary devices and personal protective equipment for each activity are available.
- The prescribed maintenance intervals and the relevant regulations are complied with.
- Installation, electrical connection and commissioning of the product is only carried out by qualified and trained personnel in accordance with this technical file.
- The operator must ensure appropriate use of the product.

2. General

The WSL & WDL Off-Circuit Tap changer are designed like a cage without oil compartment and can be installed vertically on the transformer tank cover through flange directly.

As per different operating modes there are three different types: motor drive, manual drive and top hand wheel.

As per different internal structures, and regulating mode there are six different types: Linear regulation, Single –bridging regulation, Double-bridging regulation, Y-D change-over regulation, Serial-parallel regulation, and Reversing regulation. Please find connecting detail in Fig. 3.

As per different sizes, there are four different types: A, B, C, D and E. There are two types of flange mounting: tank cover and bell type. This operating instructions contains all operating and installation materials of the Tap changer.

This instruction includes all necessary information for installing of the tap-changers.

**NOTE!**

Special design is available. Please contact Huaming for further information.

General view

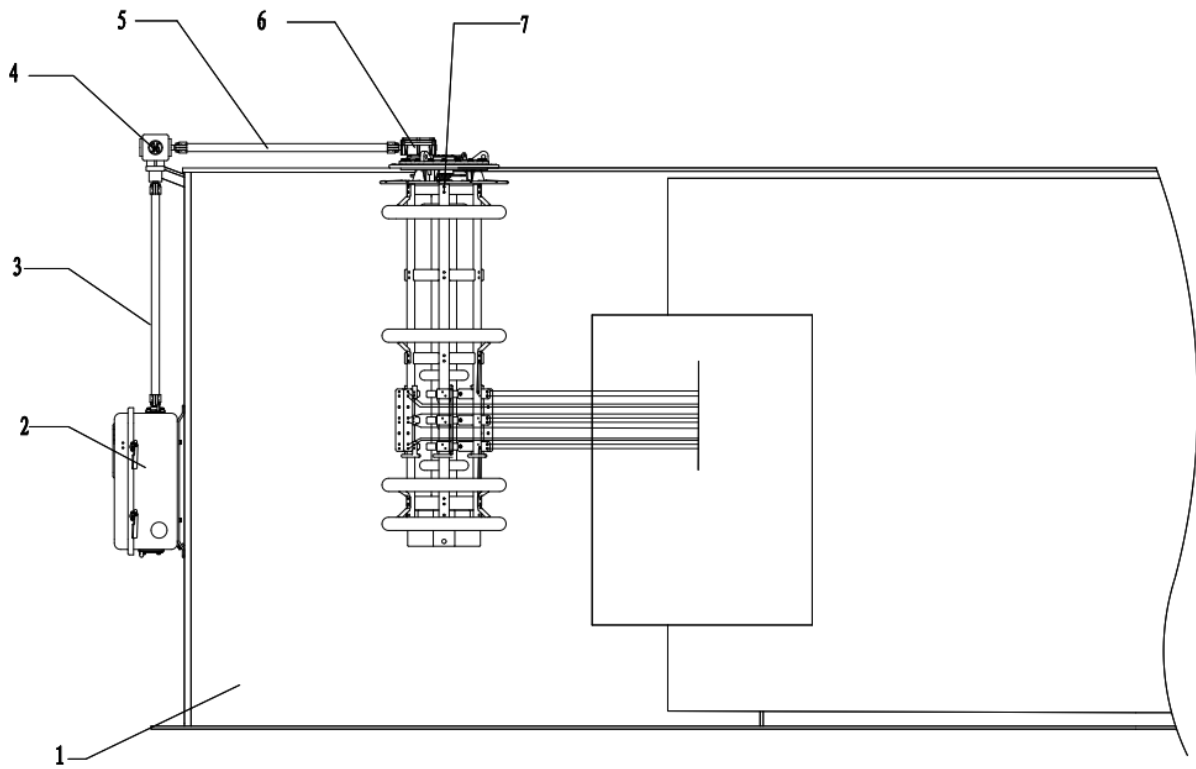


Figure 1 Overview of tap-changer inside transformer

1. Transformer tank	2. Motor drive unit
3. Vertical drive shaft	4. Bevel gear box
5. Drive shaft	6. Reduction mechanism
7. Grooved wheel mechanism	

Table 5-1 List of all parts

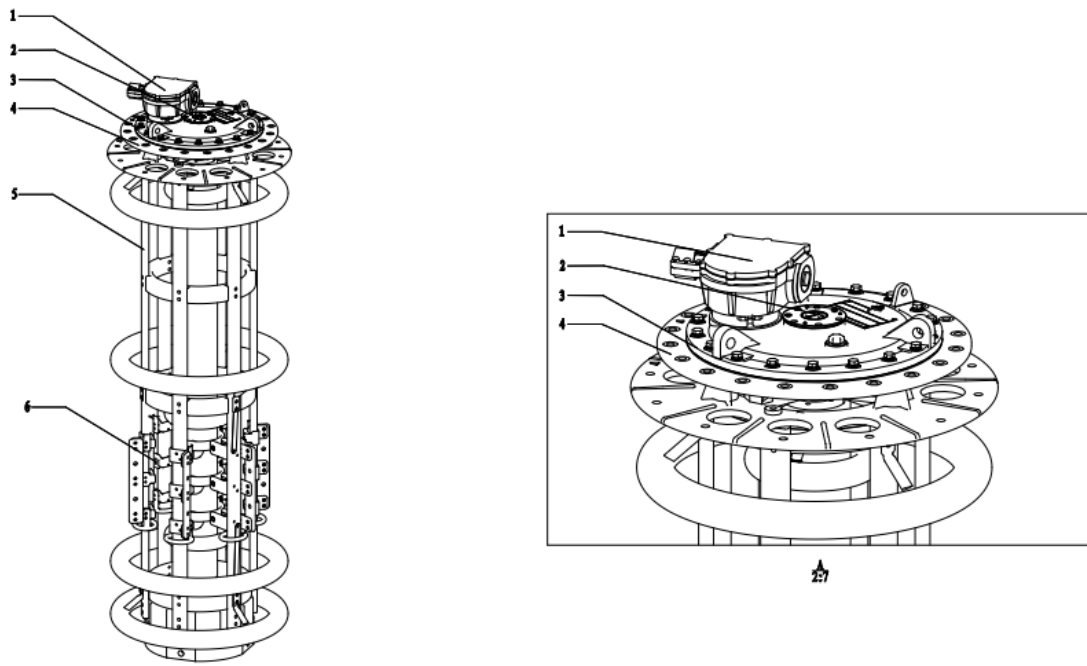


Figure 2 Overview of tap-changer

1. Reduction mechanism	2. Rupture disk
3. Tap-changer head cover	4. Tap-changer head
5. Support slats	6. Change-over selector

Table 5-2 List of tap-changer components

3. Technical data

WSL&WDL tap-changer complies with IEC 60214-1: 2014 standard.

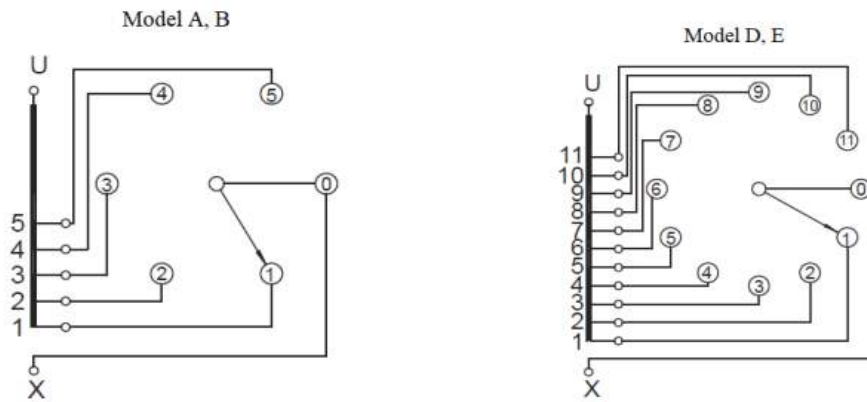
3.1 Type WSL&WDL Off-Load Tap Changer Technical Data

Item	Model		WSL、WDL							
1	Number of phases		Single-phase or three-phase							
2	Maximum rated current through (A)		600	800	1000	1200	1600	2000	2400	3000
3	Short-circuit current test(kA)	Thermally stable (3s)	9	12	15	15	20	24	26	30
		Dynamic stability (peak)	22.5	30	37.5	37.5	50	60	65	75
4	Rated frequency (Hz)		50or60							
5	Insulation level (kV)	Maximum equipment voltage	12	72.5	126	145	170	252		
			Single-phase or three-phase							
		Rated lightning surge , withstand voltage (1.2/20μs)	75	325	550	650	750	1050		
		Rated external applied withstand voltage(50Hz,1min)	35	140	230	275	325	460		
6	Internal insulation level (kV)		Table 4							
7	Size classification		A type φ350, B type φ500, D type φ600, E type φ750							
8	Maximum number of working tap positions		Type A max. working position ≤ 5, Type B max. working position ≤ 5, Type D max. working position ≤ 11, Type E max. working position ≤ 17							
9	Mechanical life		Manual switch > 20,000 cycles, electric switch > 100,000 cycles							
10	Weight (kg)		A Type				B Type、D Type、E Type			
			max:100				max:195			

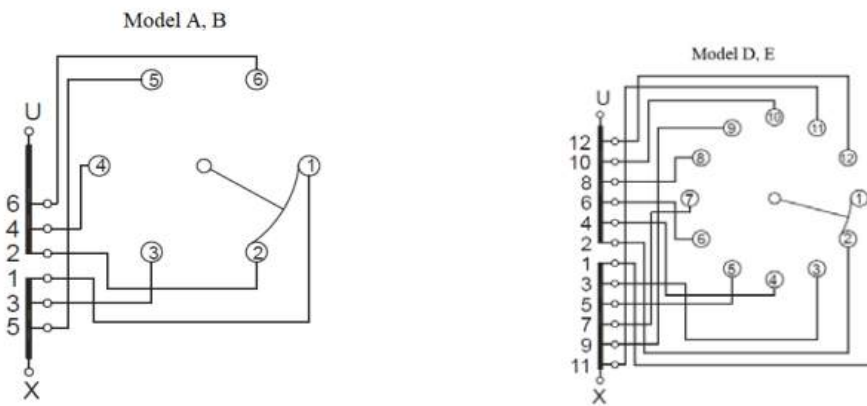
Table 6-1 WSL&WDL Technical data of the WSL series without excitation tap-changer

3.2 Principle diagram

Linear regulation



Single-bridging regulation



Double-bridging regulation

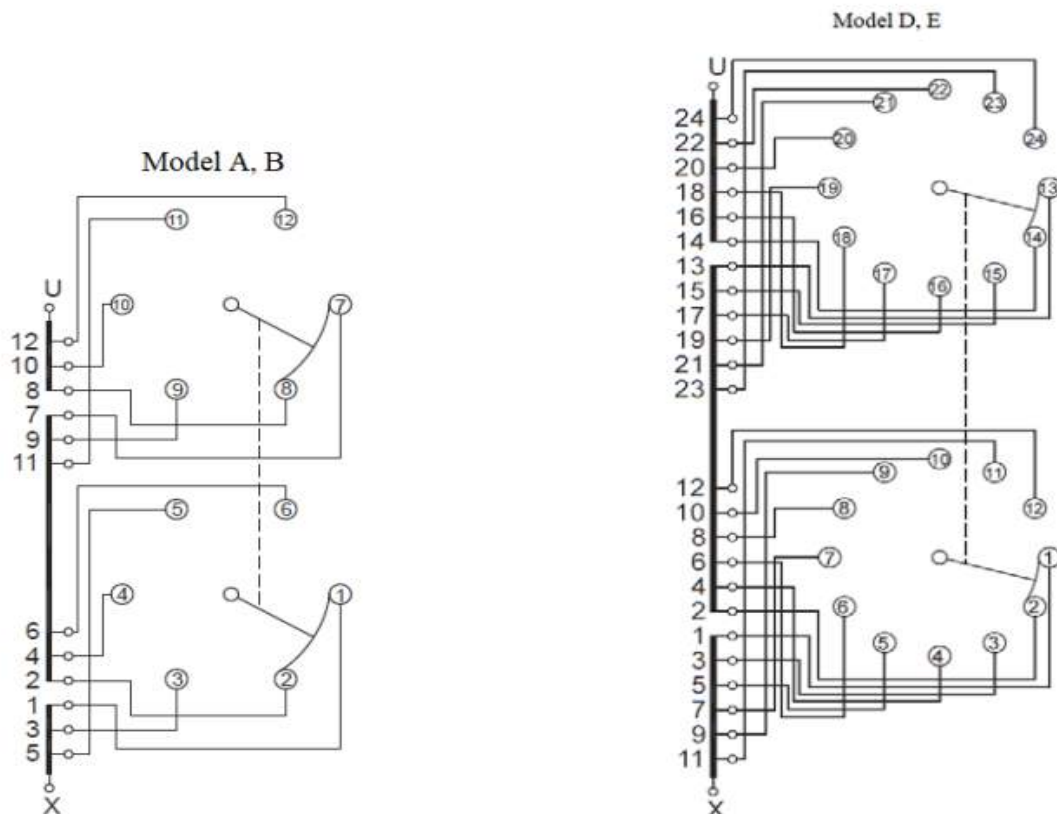
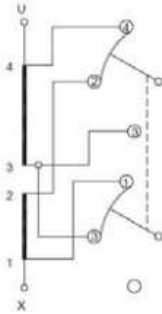
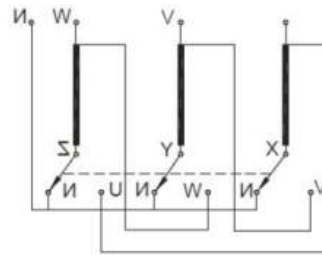


Fig. 3(a) Basic connection diagram

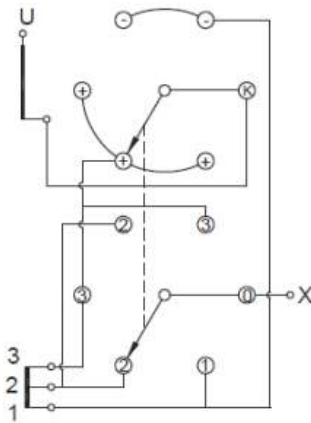
Serial-parallel transform



Y-D transform

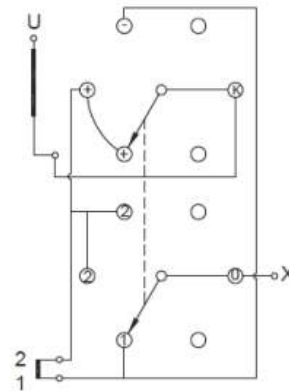


Reversing



Tap position	1	2	3	4	5
Position for regulation	+2	+1	0	-1	-2
Connection mode	$\text{K} \rightarrow +$	$\text{K} \rightarrow +$	$\text{K} \rightarrow +$	$\text{K} \rightarrow -$	$\text{K} \rightarrow -$
Connection mode	$\text{O} \rightarrow 1$	$\text{O} \rightarrow 2$	$\text{O} \rightarrow 3$	$\text{O} \rightarrow 2$	$\text{O} \rightarrow 3$

WSLII-XXX/XX-6×5



Tap position	1	2	3
Position for regulation	+1	0	-1
Connection mode	$\text{K} \rightarrow +$	$\text{K} \rightarrow +$	$\text{K} \rightarrow -$
Connection mode	$\text{O} \rightarrow 1$	$\text{O} \rightarrow 2$	$\text{O} \rightarrow 2$

WSLII-XXX/XX-4×3

Fig. 3(b) Basic connection diagram

3.3 Operation conditions

- 1). The operating temperature of the tap-changer in oil is $\geq -25^{\circ}\text{C}$ and $\leq 105^{\circ}\text{C}$.
- 2). The storage environment temperature of the tap-changer is $\geq -25^{\circ}\text{C}$ and $\leq 40^{\circ}\text{C}$, and the humidity should be not higher than 85%. If beyond this standard, please mark your requirements before ordering. The tap-changer material and accessories will be specially processed to meet the special requirements of the environment.
- 3). After being mounted to the transformer, the non-perpendicular degree between the tap-changer and the ground should be less than 2%.
- 4). No sever dust and other explosive or corrosive gases on the installation site.

**Prohibition!**

Do not store the tap-changer or motor drive unit in a place with high humidity ($\geq 85\%$). This will cause condensation.

**NOTE!**

Special design is available to meet particular environment conditions. Please contact Huaming for further information.

4. Type designation

4.1 Type designation

WSL tap-changers have a variety of specifications due to different combinations of the number of phases, the maximum rated through current, the highest voltage of the equipment, the insulation class of the selector and the connection mode. Therefore, the above-mentioned performance parameters should be clearly reflected in the model of the tap-changer, and the detailed description of each code is shown in Figure 4.

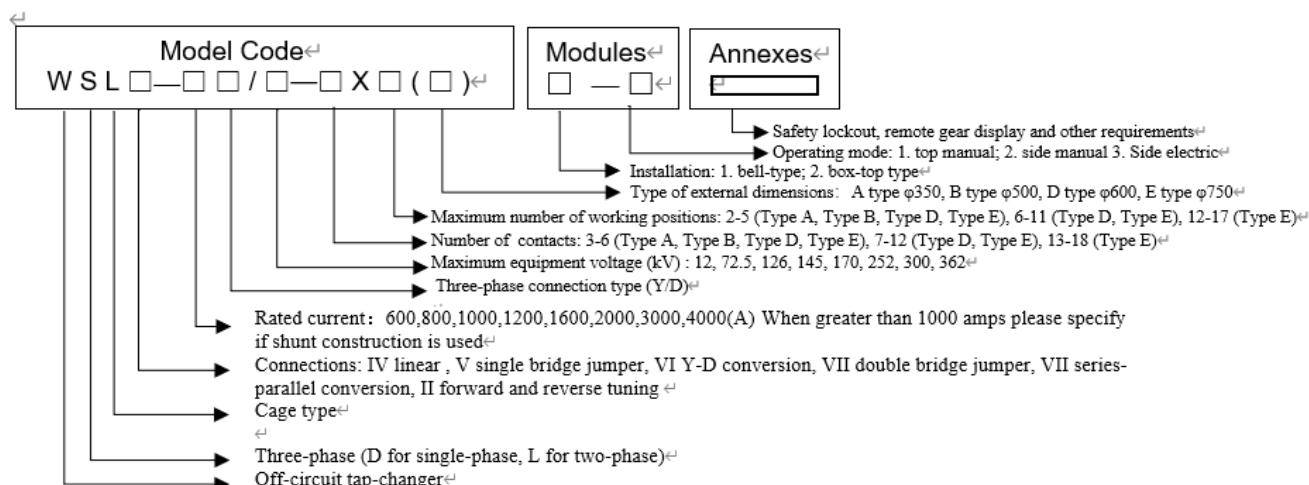


Figure 4 WSL&WDL Tap changer model designation

4.2 Basic connection

Table 6 Tap changer coupling types and their designations

Designator	IV	V	VI	VII	V III	II
Connection method	Linear	Single bridge jumper	Y-D conversion	Double bridge jumper	Series - parallel conversion	Forward and reverse

4.3 Voltage regulation

No-load voltage regulation, when the load is rated voltage load, it should be consistent with the grid voltage, usually in the middle tap.

For example: the intermediate tap of the 9-level linear adjustment should be in the 4th-6th tap position.



WARNING

By default, the tap changer stops at the rated tap position (intermediate tap position). Make sure that the transformer is powered off before adjusting the tap position.

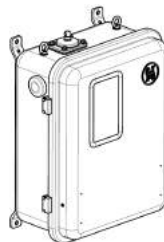
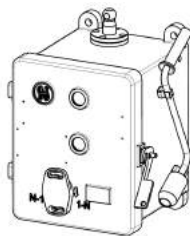
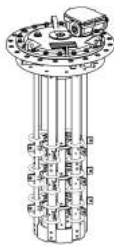
5. Packing, transportation & storage

5.1 Supplied components

The complete set of tap-changer contains below parts

- Tap changer main body
- For ground motor driving type tap changer, the tap changer will be delivered with Motor drive Unit SHM-D, bevel gear box, vertical and horizontal drive shafts
- For ground manual driving type tap changer, the tap changer will be delivered with manual driving device, bevel gear box, vertical and horizontal drive shafts

Please refer to the corresponding packing list delivered with the tap-changer for exact supplied components.



5.2 Transportation & acceptance

5.2.1 Transportation

- Packings is suitable for all common kinds of transportations. The boxes of tap selector and MDU are allowed to be appropriately stacked during storage, and the load-bearing capacity of the box's top surface shall not exceed 500kg/m².

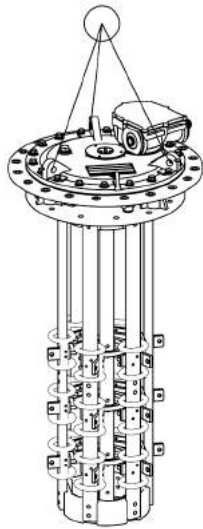
- Boxes should be placed reasonably according to the position of the gravity center. When lifting it up, the markings on the box must be observed (Figure 5).



Figure 5 Example of markings on the box

5.2.2 Lifting

The lifting of the OCTC body after unpacking is shown in the below figure:



The octc is designed with lifting holes. When lifting, all lifting holes need to be completely fixed and then hoisted at a constant speed and stably.



WARNING

Before lifting, it is necessary to check whether the sling is completely fixed, and pay attention that the lifting angle does not exceed 45°.

The lifting process should be uniform and stable, avoid sudden acceleration of lifting and lowering, and prohibit high-speed lifting and rapid landing



DANGER!

When lifting and storing the boxes, people are not allowed to stand under or near the boxes in case the boxes drop. All boxes must be stored properly and safely.



NOTE!

Choose appropriate forklift and crane according to the dimension and gross weight of the boxes.

**NOTE!**

Danger of damage to the tap-changer due to falling and tipping over!

- Lifting must be carried out by a professional or designated person
- The hoisting capacity of the equipment must be greater than 500kg.
- Do not stand within a radius of one meter below the hanging load while lifting.

5.2.2 Goods receiving

Each batch of delivered goods must be checked by the consignee before accepted (signing acceptance confirmation). Following items must be checked by the consignee carefully:

- When receiving the goods, check the quantity of components received according to the packing list.
- Check whether the packing is intact. If it's found to be damaged (if possible, take pictures of the box and damaged goods) or the information of the goods does not match the packing list, consignee should refuse to sign for acceptance, and immediately report the problem to the shipping company and Huaming.
- The above steps are also applicable to corrosion of the goods due to moisture immersion (rain, snow, condensed water). The components must be stored in a dry place before installation.
- The tap-changer must be stored in a sealed packing tank and opened before installation only.



NOTE!

When receiving the goods, please check carefully the packing and the supplied components according to the packing list. In case anything is broken or missing, please contact Huaming in time.



NOTE!

The transportation and lifting of the box must be operated by professionals. The carrying capacity of transportation vehicles and cranes should be more than 1000kg.



NOTE!

The packing box should be firmly fixed during transportation and lifting, and be handled with care to avoid possible damage to the product due to vibration, impact, falling, tipping, bumping, shaking, etc.



NOTE!

If the box drops or is severely impacted, the possibility of damages must be considered. Comprehensive inspection of the product should be carried out, and if necessary, inspection(repair) shall be done by the Huaming's professional and technical personnel.

5.2.3 Acceptance

Unpack the box and look for damage during transportation.

- Transport the box to the installation site.
- Check the equipment and its accessories according to the packing list.



NOTE!

During unpacking and inspection, avoid damaging the original package.



NOTE!

If the received goods and its accessories do not match the list, contact Huaming in time.

5.3 Storage

When the goods are stored continuously for more than 1 year, the equipment should be inspected thoroughly before installation.

Specially packed goods with sealed packing can be stored outdoors if they meet the following conditions.

- The storage environment temperature should not be $\geq -25^{\circ}\text{C}$ and $\leq 40^{\circ}\text{C}$.
- The tap-changer must not be stored in an environment with flammable, explosive and corrosive gases.
- It must be protected from moisture, dust, rodents and ants.
- Regularly check whether the stored equipment is abnormal
- If the tap-changer is stored for a long time, the desiccant must be replaced regularly, and re-packed with sealed packing.



NOTE!

After long-term storage, the tap-changer and its accessories must be carefully inspected before installation.

6. Installation

6.1 Pre-installation check

6.1.1 Check that the specifications of the tap-changer correspond to the design requirements of the transformer and that the technical documents such as the certificate of conformity, operating instructions and packing list are complete when delivered.

6.1.2 Check that the components of the tap-changer are free from defects and deformations.

6.1.3 Operate the tap changer for one cycle and check that there is no jamming of the movement or damage to the parts and that the working position of the contacts is the same as that indicated by the position indicating device.

6.1.4 Measure the contact resistance of the contacts in each tap position to ensure that it corresponds to the data on the certificate of conformity.

6.2 Installation

This type of tap changer does not contain oil compartment and can be directly mounted into the transformer oil tank. Corresponding to two different box structures of the transformer, the tap HM0.460.602 19 changer also has two different installation methods as follows:

6.2.1 Installation of the off-circuit tap changer on the tank cover type transformer (fig. 7)

6.2.1.1 Clean all the surfaces for the sealing, and put oil -proof gasket on the mounting flange;

6.2.1.2 slowly lower the tap changer into the transformer;

6.2.1.3 damage to the tap-changer terminals and screen cover should be avoided during lifting.;

6.2.1.4 Mount the tap changer into transformer installation flange after confirming the mounting position tap changer is in correct position.



DANGER!

If heavy components are not placed on a flat surface, they may tilt and cause injuries! Therefore, Place the device on a flat and horizontal floor to prevent structural damage caused by possible tilt and uneven force.

6.2.2 Installation of the off-circuit tap changer in bell type transformer (fig. 8)

6.2.2.1 Place the tap changer vertically and dismantle the head flange.

6.2.2.2 Remove the three hex screws which connect between the middle flange and the supporting flange. Take out the middle flange; make sure to keep all the demounted parts.

6.2.2.3 Lift the tap-changer and place the supporting flange on the temporary support frame of

the transformer, adjust the relative position of the supporting flange to the head flange and fix it. If the relative position of the supporting flange and the head flange is difficult to calibrate, the bell-type cover can be pre-assembled to the transformer body for adjustment.

**DANGER!**

If heavy components are not placed on a flat surface, they may tilt and cause injuries! Therefore, Place the device on a flat and horizontal floor to prevent structural damage caused by possible tilt and uneven force.

6.2.2.4 Pre-assembly Process:

- a) Lift the bell type box cover of the transformer, cover the transformer, clean the sealing surface and install the sealing gasket.
- b) Fix the intermediate flange on the transformer mounting flange.
- c) Use the two lifting rings of support flange to lift the tap changer, adjust the set position, fix the tap changer on the middle flange, clean the sealing surface, put on the sealing gasket,
- d) Finally install the head flange.

**Note:**

The three red triangle marks on the support flange, middle flange and head flange shall be in a straight line (Appendix 23 &24)

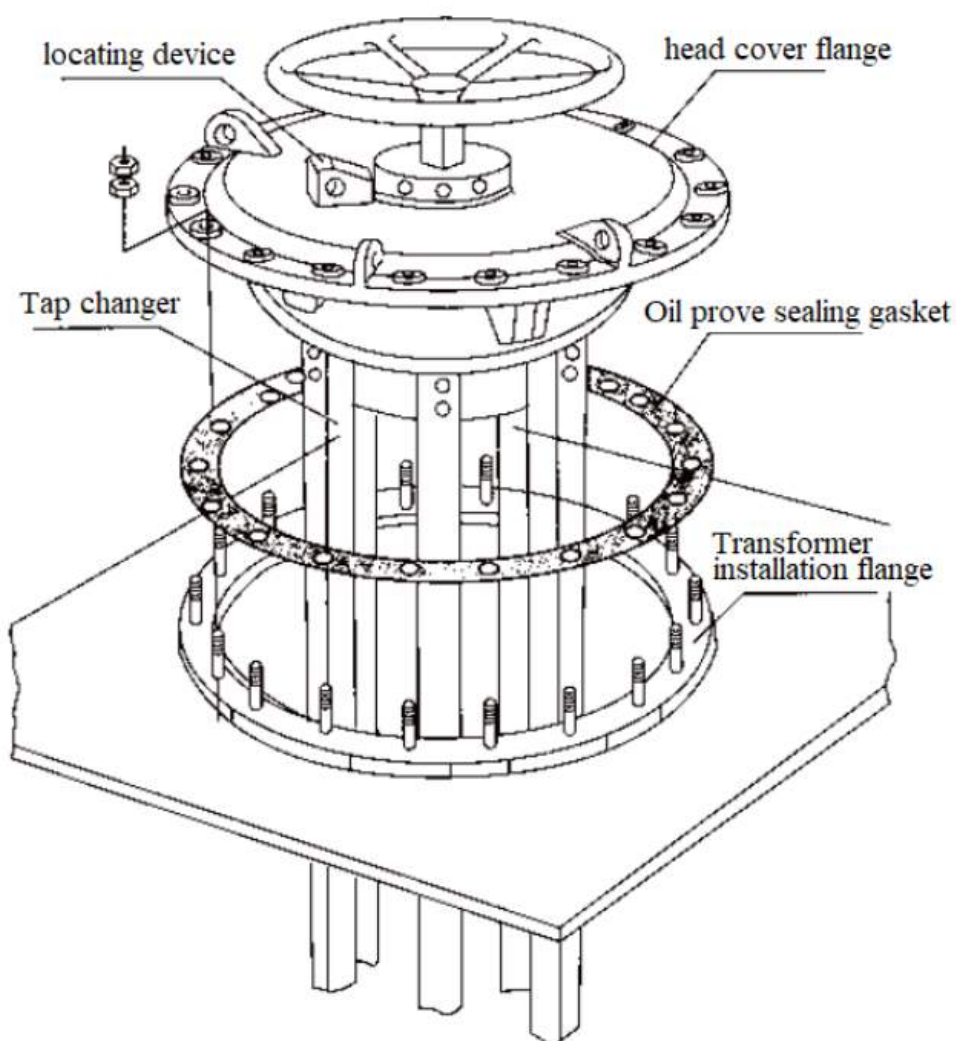


Figure 7 Installation diagram of the tap changer on the tank cover type transformer

- (a) All of the tap leads should be carefully handled and properly fastened.
- (b) There should be no pulling force between any tap leads and the tap changer.
- (c) The leads between flange of the tap changer cover and the transformer cover are to be grounded

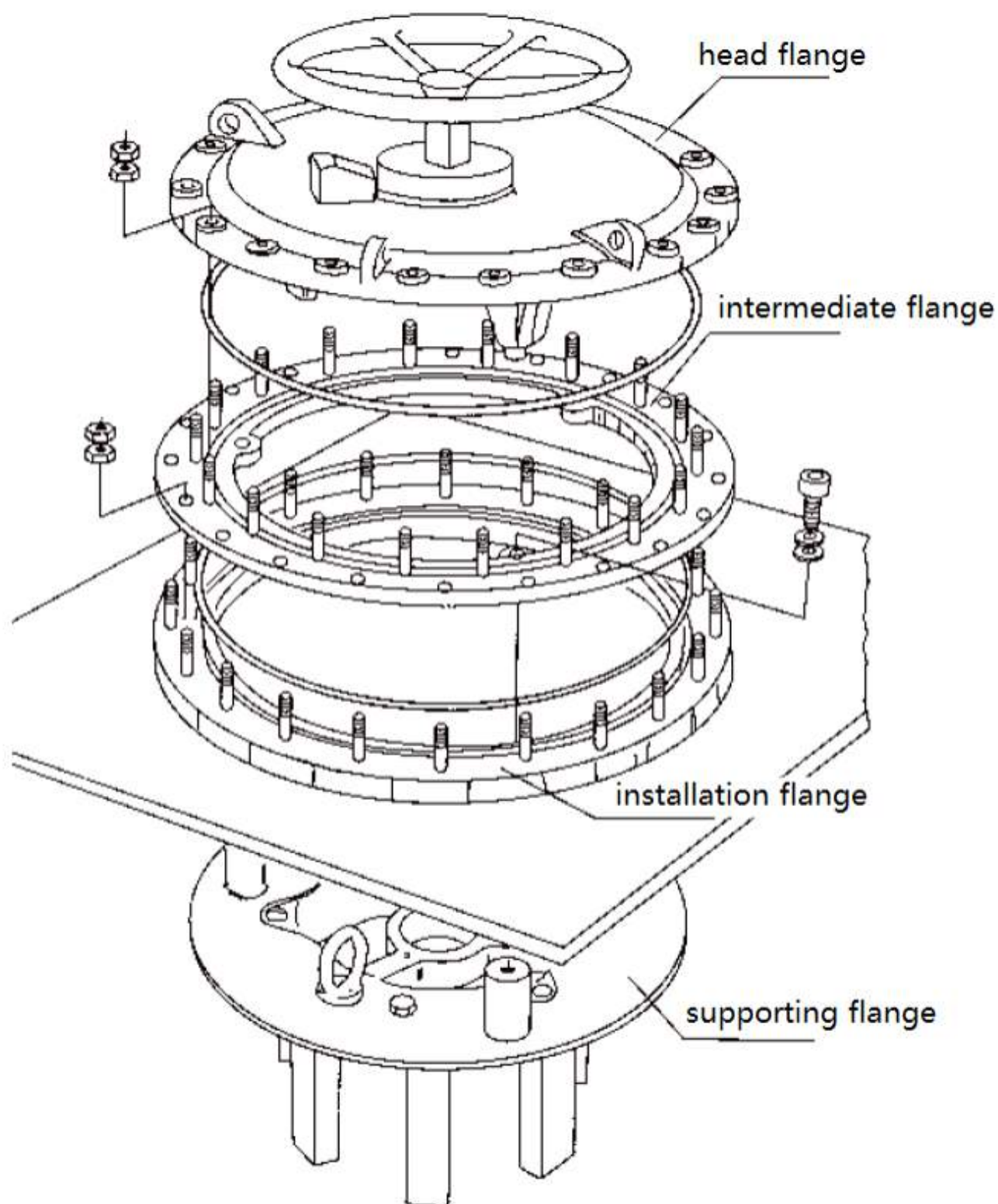


Fig. 8 Installation diagram of the tap changer mounting on the bell type transformer

6.2.3 Mounting of the motor drive unit or the manual drive unit



PROHIBITION!

The transformer cannot be energized until the drive unit and the off-circuit tap changer are in the same operating position. That is, the connection between the tap changer and the motor drive unit must be checked before energizing the transformer.

6.2.3.1 T

support plate protruding from the top of the transformer, pay attention to the alignment of the horizontal output shaft of the bevel gear box and the output shaft of the gear box at the head of the tap changer, determine the size of the transmission shaft, leave a gap of about 2mm. Connect the bevel gear box with the gear box on the flange of the tap changer head after the length of the square shaft is processed. Pay attention to adjust the horizontal position, and make the gear box, the transmission shaft and bevel gear box are in a straight line.

6.2.3.2 Install the motor drive unit or the manual unit onto the side tank of the transformer.



CAUTION!

The surface of the installation box must be flat. The output shaft of the mechanism must be vertical to the ground, and should be aligned with the vertical output shaft of the bevel gear box on the support plate at the top of the transformer. Check the size of the transmission square shaft, and leave a gap of about 2mm. Connect the motor mechanism or manual mechanism with the bevel gear box after the length of the square shaft is processed, and turn the locking plate 90 ° after fastening the connecting screws, so as to prevent loosening.

6.2.4 Connection of the motor unit and the tap changer

6.2.4.1 Working position indicated by the motor drive unit must be consistent with the tap changer's, and then connect horizontal and vertical shafts.

6.2.4.2 Manually operate the motor drive unit to record the number of turns of the tap changer in two directions as follows: Manually operate the motor drive unit in one direction until the moment of red arrow in the center of the head flange of the tap changer aligns with the number and not move, start to record the number of manual turns, shake until the center line (red) in the green area on the tap changing indication wheel is aligned with the center of the digital window, the number of turns shaken is m, reverse the operation with the same method and record the number of turns as n (see Figure 9).

6.2.4.3 Adjustment: if $m-n$ or $n-m \leq 3.75$ turns, means it meets the requirements and no need adjustment. Conversely, adjust as follows: disconnect the vertical shaft from the motor drive unit, then add 3.75 turns to the hand-operated motor drive unit in the direction with more turns, and reconnect the vertical shaft to meet the requirement of " $m-n$ or $n-m \leq 3.75$ ".

6.2.4.4 Measure the transformer's transforming ratio at each tap position.

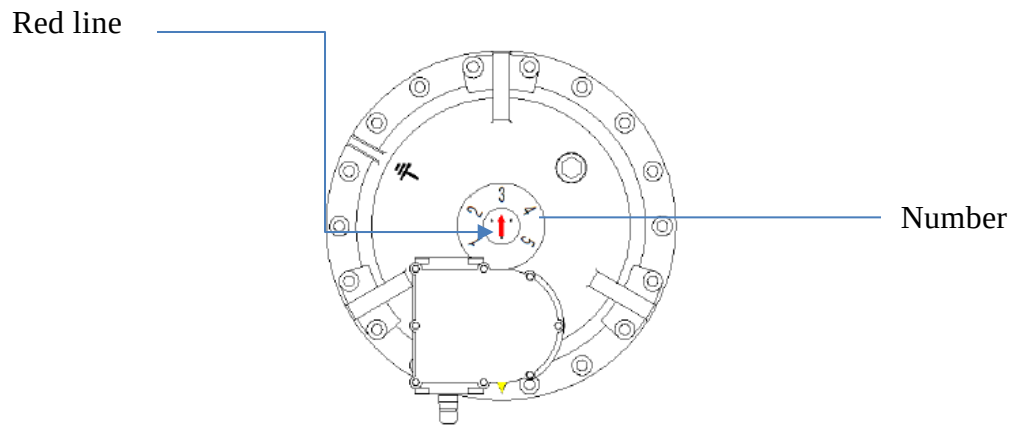


Diagram for position indicator on head flange of tap changer

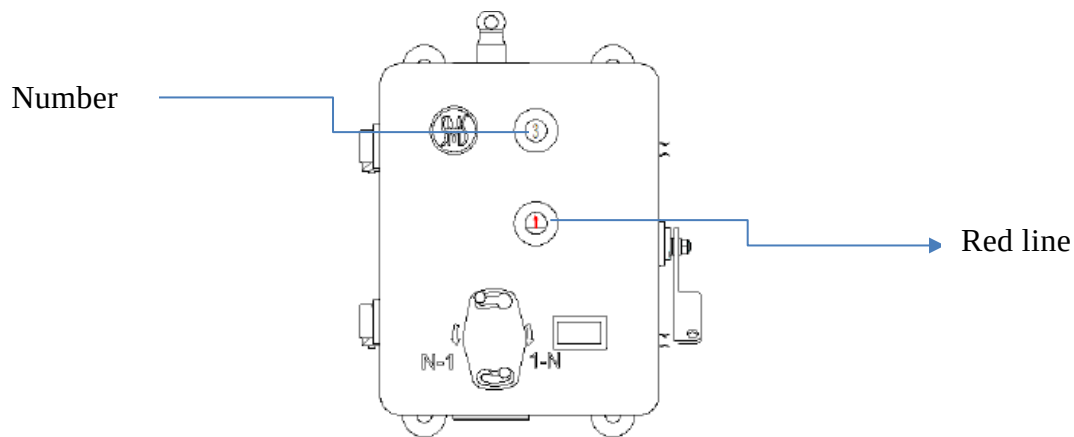


Diagram for position indicator on manual drive unit

Fig 9 Diagram of the gear display



NOTE!

1. Do not operate while the transformer is energized.
2. Make sure the door always closed and locked of tap changer when the tap changer is in operation.
3. Operate tap changer from outside of the control cabinet and make sure both indicators are in the same position.

7. Transformer ratio test

It is suggested that the ratio test should be carried out before drying.



NOTE!

Material damage

Operation without oil will lead to damage of OCTC.

- Without oil, tap-changer cannot be operated more than 10 cycles before drying.
- For the first operation of tap -hanger after transformer being dried, make sure tap selector is fully immersed in oil and the diverter switch oil



DANGER!

Incomplete tap change operation can cause damage to the OCTC!



NOTE!

It is suggested that the ratio test should be carried out before drying.



NOTE!

For bell-type tap-changer, head flange, diverter switch and top cover need to be assembled temporarily.

Check each operating position on the inspection window against the included connection diagram

After completion of the test the OCTC has to be operated back to the adjustment position.

8. Drying process and oil filling

8.1 Drying process

The tap changer is to be dried with the transformer; the drying temperature and time are the same as that of the transformer. The electric insulation level of the tap changer can only be guaranteed after the drying procedure. In order to guarantee the electric insulation level of the tap changer, the tap changer should be dried with the transformer, and the drying process is same as the transformer.

**PROBITION!**

Do not operate the tap changer without oil after drying. If necessary, smear all contacts with some transformer oil before operation

**NOTE!**

The tap changer should be immersed into the transformer oil immediately after the drying process.

**NOTE!**

Loosen the bleeder on the head flange of the tap changer to release gas during oil refilling of the transformer tank. The transformer oil should strictly adhere to the standard, especially regarding the insulation level and the water percentage level.

8.2 Oil filling

After drying, in order to prevent too much moisture from entering the oil compartment, the oil compartment (diverter switch insert already installed) should be refilled with oil as soon as possible.

The tap-changer body does not contain an oil compartment and needs to be mounted on the transformer and filled with oil via the transformer oil pipe

9. OCTC put into operation on site

The off-circuit tap changer can only be operated when both high voltage side and low voltage side of the transformer are de-energized.

9.1 Operation of top cover tap changer :

9.1.1 Loosen the positioning screw on the tap changer positioning device (Fig.10)

9.1.2 Turn the hand wheel and observe whether the arrow is aligned with the red arrow on the indicator from the observation window. The alignment indicates that the gear change is successful. (Fig.11)

9.1.3 Tighten the positioning screw into the positioning hole to complete the tapping action change.



Danger

Danger of death and severe injury from electrical voltage..

- Switch off transformer on high and low voltage side before maintenance.
- Lock transformer on high and low voltage side before maintenance to prevent it being switched back on.
- Short-circuit transformer on high and low voltage side before Operation .
- Check the newly set operating position using the tap position indicator after Operation and before switching the transformer on.



Material damage

Damage to the tap-changer due to damaged Assembly .

- Damage to property and the environment.
- Electric voltage will damage the off-circuit tap-changer and transformer.
- Switch off transformer on high and low voltage side before Operation.
- Lock transformer on high and low voltage side before Operation to prevent it being switched back on.
- Short-circuit transformer on high and low voltage side before Operation.
- Check the newly set operating position using the tap position indicator after Operation and before switching the transformer on.

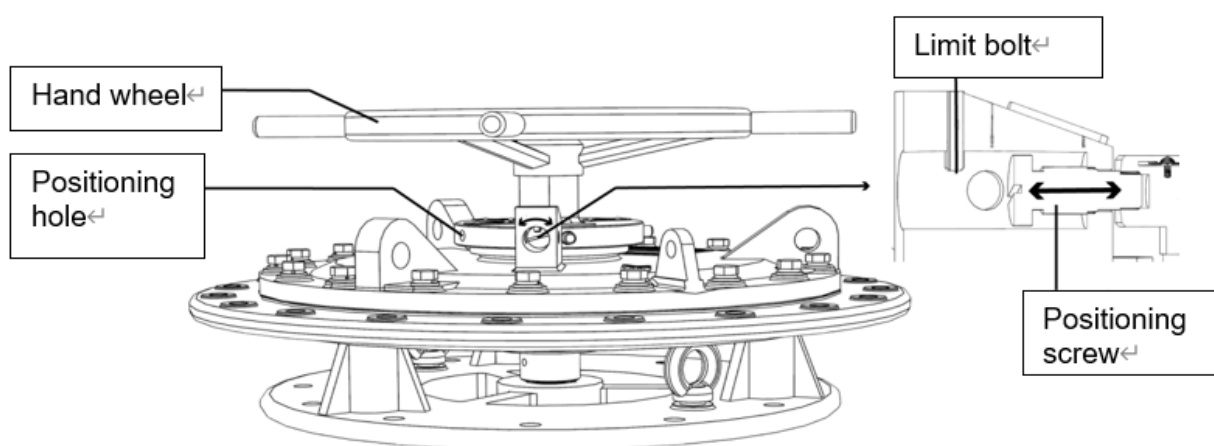


Fig10-Switch the positioning device by tightening the positioning screw

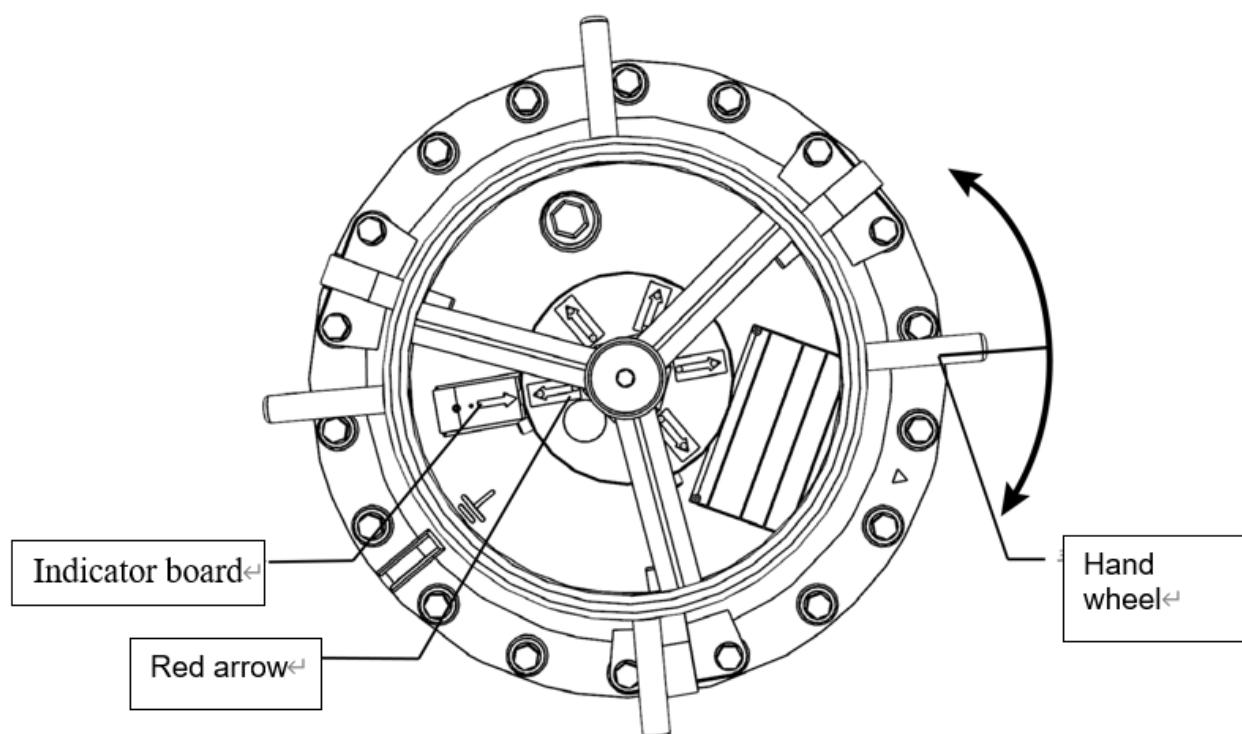


Figure.11 Turn the hand wheel and confirm that the indicator board is aligned with the red arrow

9.2 Operation of floor manual type tap changer:

9.2.1 Open the cover of the hand wheel on the manual box, take out the positioning piece, and insert the hand wheel (see Figures 12 and 13)

9.1.2 Turn the hand wheel and observe from the observation window whether the arrow and the red arrow of the indicator are aligned. Alignment means that the gear is successfully changed. (See Figure 14)

9.1.3 Pull out the handle, cover the handwheel cover and insert the positioning piece to complete the tapping action change.

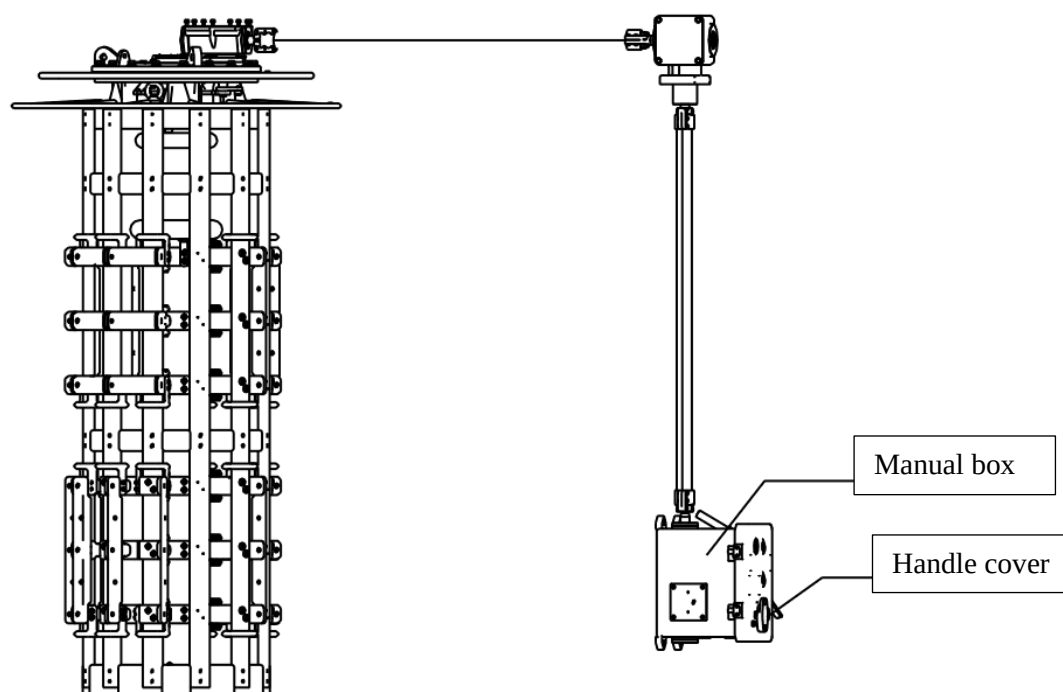


Figure 12 Schematic diagram of the position of the manual box and the position of the handle cover

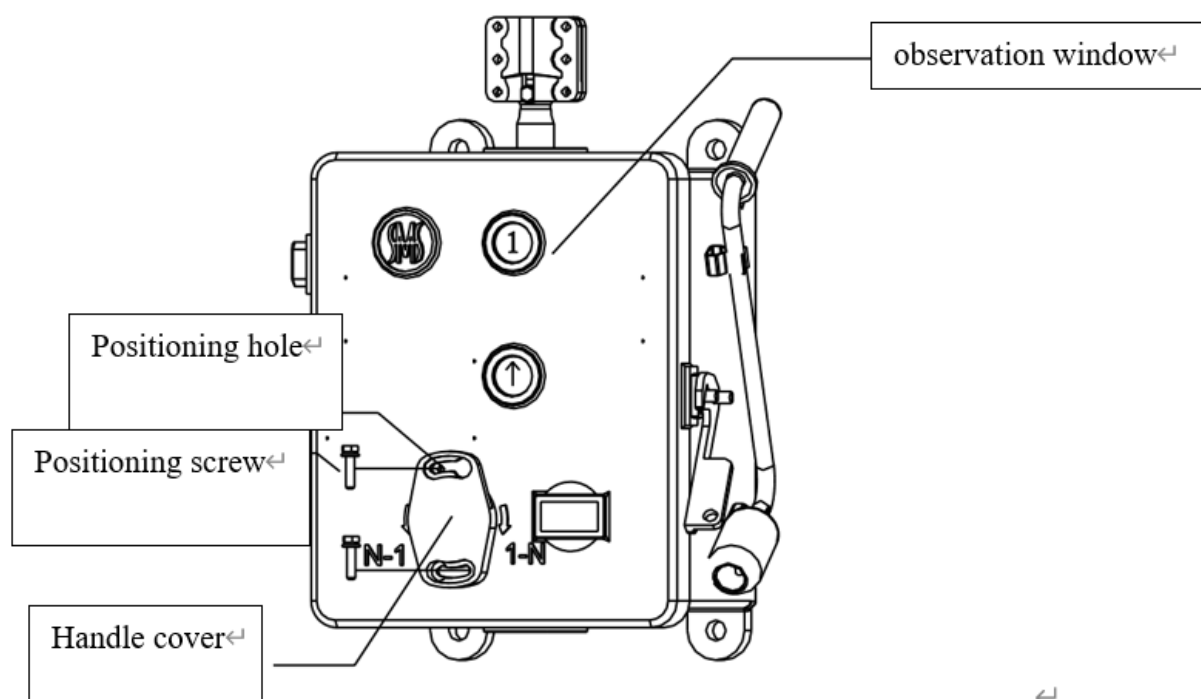


Figure 13 Take out the positioning piece and remove the handle cover

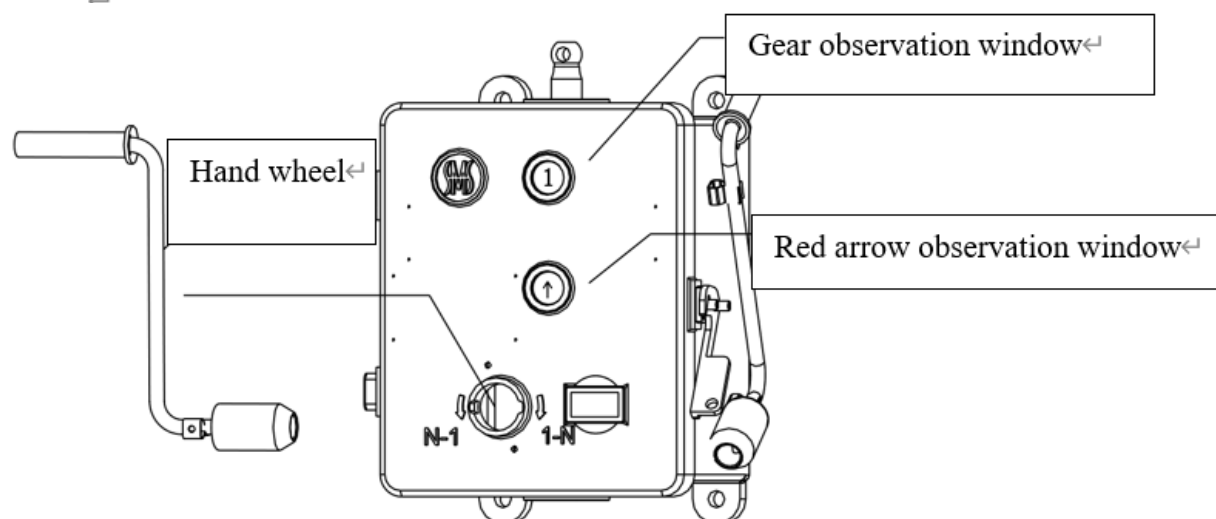


Figure 14 Turn the handwheel and confirm that the sign is aligned with the red arrow

10. Maintenance



Electric shock!

An energized transformer may cause death or serious injury.

- Turn off the high voltage and low voltage ends of the transformer.
- Lock the transformer to prevent accidental restart.
- Ensure that all components are powered off.
- Clearly ground all transformer terminals (ground wire, ground isolation switch) and short-circuit them.
- Cover or isolate adjacent live parts.



Electric shock!

Operation of energized tap-changers may cause personnel death or serious injury.

- Disconnect all auxiliary circuits, such as tap change monitoring devices, pressure relief valves and pressure monitors.
- Ensure that all components are powered off.



Warning!

Risk of explosion!

Explosive gas in the oil compartment of the tap-changer, transformer, piping system, oil conservator, and the outlet of the moisture absorber may ignite or explode, causing serious injury or death.

- Ensure that there are no open flames, hot surfaces, or electric sparks (for example, electric sparks caused by static electricity accumulation) in the environment around the transformer, and that the above situations will not occur.
- Do not operate any electrical equipment (for example, impact wrenches have the risk of sparking).
- Only use conductive and grounded hoses, pipes and pump operation equipment suitable for flammable liquids.



Check the motor drive unit!

Condensation in the MDU protection case can cause damage to MDU.

- Always keep MDU's protection case sealed.
- If the operation is interrupted for more than 2 weeks, connect and operate the anti-condensation heater in the MDU. If this requirement cannot be met (for example, the heater cannot be energized during transportation), a sufficient amount of desiccant must be placed inside the case.

10.1 Inspection

10.1 Before inspection

During the inspection and maintenance operations, it is necessary to wear high-voltage insulating clothing, high-voltage insulating boots, high-voltage insulating gloves, and high-voltage electroscope.

Maintenance operator must be professionally trained and held by electrician's license holders.

Transformer must be de-energized and all terminals shorted and grounded before servicing.

Interval	Description	Instruction
Once per year	Check the door tightness, cable bushings, MDU protective cabinet and control panel's ventilation device	Visually observe whether the sealing parts are aging, damaged, and whether oil seeps. If yes, replace with new ones immediately. The MDU and control panel ventilation is checked by breather test. If it is damaged, replace with new one immediately.
Once per year	Check the tightness of tap-changer head cover, protective relay and all pipes' connectors.	Visually observe whether the sealing parts are aging, damaged, and whether there is oil leakage. Then, use the insulating glue sticks to press the sealing parts to check whether they have elasticity. If the sealing parts are aging and damaged, they should be replaced immediately.
Once per year	Check the MDU's protective cabinet and if the heater inside the control panel works normally or not.	Visually check whether the circuit components are damaged with black and yellow pins. If there is any damage, please contact the after-sales service. Then, open the temperature controller and conduct a temperature control test. Use a thermometer to check whether the setting temperature is reached.
Once per year	Check whether the silica gel desiccant in the transformer oil conservator is working properly	Put the humidity indicator card into the oil reservoir for at least half an hour. If the humidity is greater than 10%, the desiccant should be replaced, or the desiccant should be replaced every other year.
Once per year or every 20000 operations	Check the oil sample. If it doesn't meet the parameters in Table 13-2, it's recommended to change the transformer oil.	Use a syringe to sample 50-250ml at the top and bottom respectively. If the parameters in Table 8 are not met, it is recommended to replace the transformer oil.

Table 7 Inspection instruction

Item		Breakdown voltage	Water content
Limit values for insulating oil in accordance with IEC 60296			
When commissioning the transformer for the first time		> 60kV/2.5mm	< 15ppm
During operation	Class I tap-changer	> 30kV/2.5mm	< 40ppm
	Class II tap-changer	> 40kV/2.5mm	< 30ppm
After maintenance		> 45kV/2.5mm	< 15ppm
Limit values for natural esters in accordance with IEC 62770			
When commissioning the transformer for the first time		> 60kV/2.5mm	≤100ppm
During operation	Class I tap-changer	> 30kV/2.5mm	≤200ppm
	Class II tap-changer	> 40kV/2.5mm	≤200ppm
After maintenance		> 45kV/2.5mm	≤100ppm

Table 8 Oil sample limit values

Interval	Instruction
After 300,000 times OCTC operation (based on the number shown in MDU's counter OCTC inspection and maintenance. Regarding this, please contact Huaming's after-sales department.	After 300,000 times OCTC operation (based on the number shown in MDU's counter OCTC inspection and maintenance. Regarding this, please contact Huaming's after-sales department.
After 800,000 times OCTC operation (based on the number shown in MDU's counter	Change the diverter switch insert. Regarding this, please contact Huaming's after-sales department.
After 1,500,000 times OCTC operation (based on the number shown in MDU's counter	Tap selector maintenance. Regarding this, please contact Huaming's after-sales department.

Table 9 Maintenance interval

**NOTE!**

The maintenance of the tap changer shall be performed by Huaming technician or any service team authorized by Huaming, and usually the maintenance can be finished within one day.

Please write to export@huaming.com for any need of support.

10.2 Notice**NOTE!**

OCTC must be maintained strictly follow the manual to ensure its good performance. Otherwise, it will cause damages to operations of tap-changer and transformer.

**NOTE!**

If the tap-changer does not regulate the voltage for a long time or has a tap position that is not used for a long time, no less than 5 cycles of tap change operation should be performed in the full tap range during a power cut.

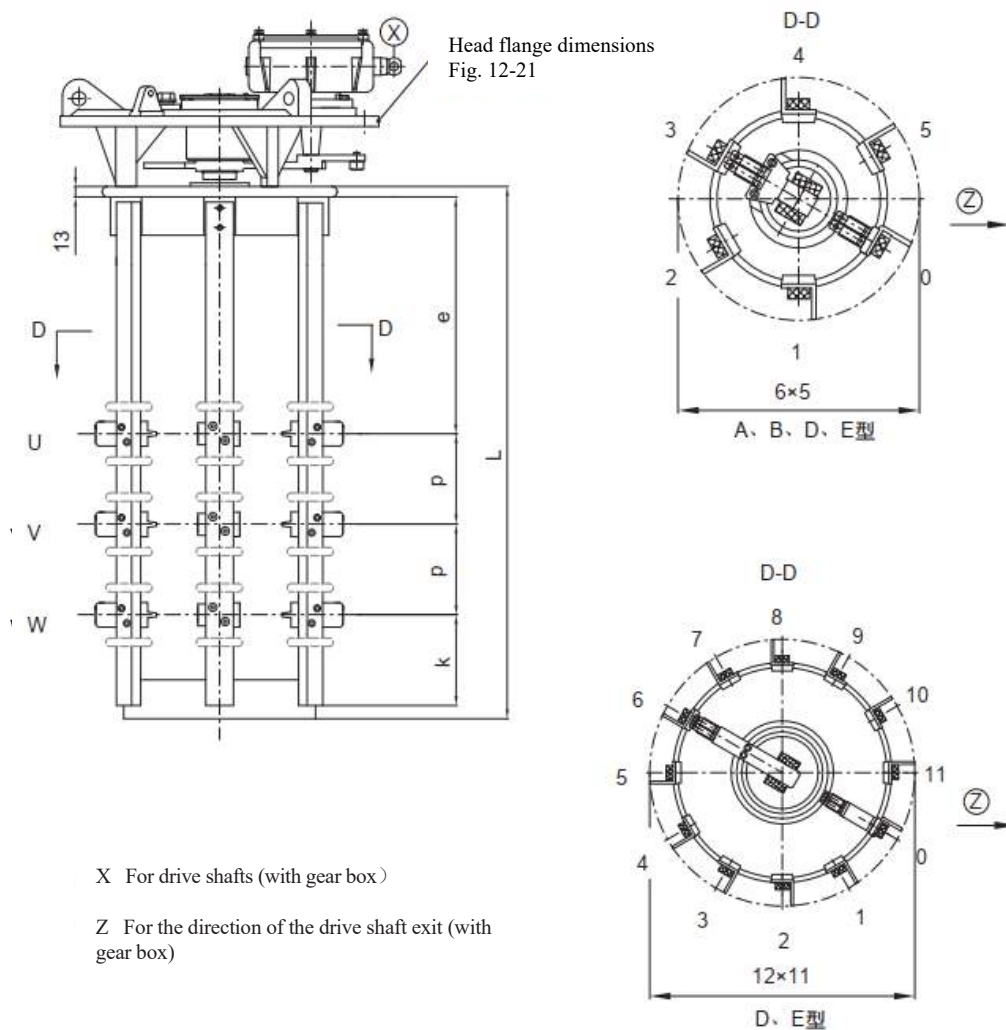
**NOTE!**

Limit values for natural esters in accordance with IEC 62770
Limit values for insulating oil in accordance with IEC 60296

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Appendix .1 600-1000A Dimensional drawing of the Linear tap-changer



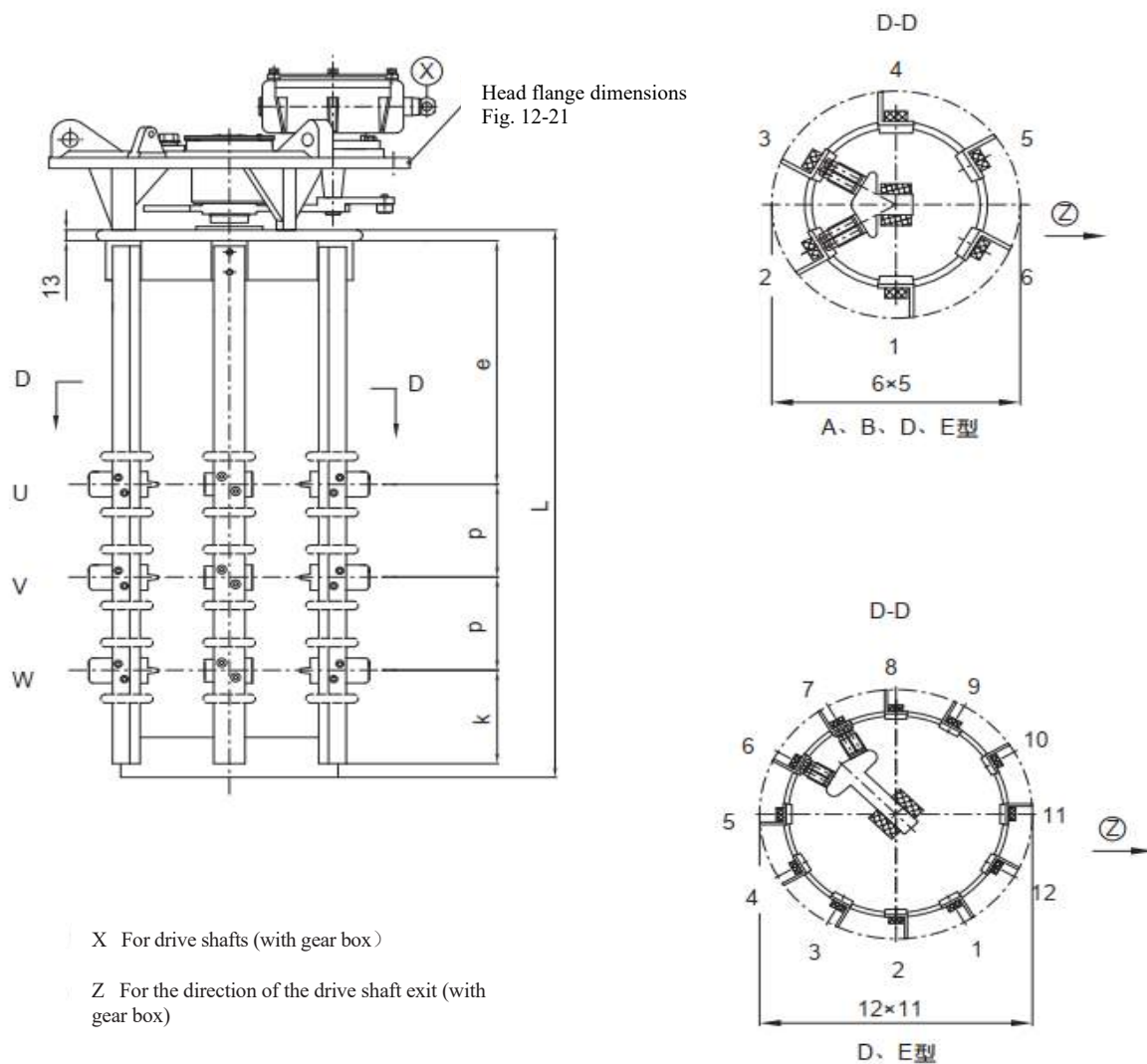
Note:

- 1 Type A only for currents ≤ 800 A
- 2 Contact size reference Fig.20
- 3 Cage contact outer diameter for types A, B, D and E respectively $\Phi 350\text{mm}$ 、 $\Phi 500\text{mm}$ 、 $\Phi 600\text{mm}$ 和 $\Phi 750\text{mm}$

Three-phase connection	Wye				Delta			
	e	p	k	L	e	p	k	L
Maximum equipment voltage								
12 kV	200	130	135	630	200	130	135	630
72.5 kV	340	130	145	780	340	280	145	1080
126 kV	470	170	155	1000	470	410	155	1480

Unit :mm

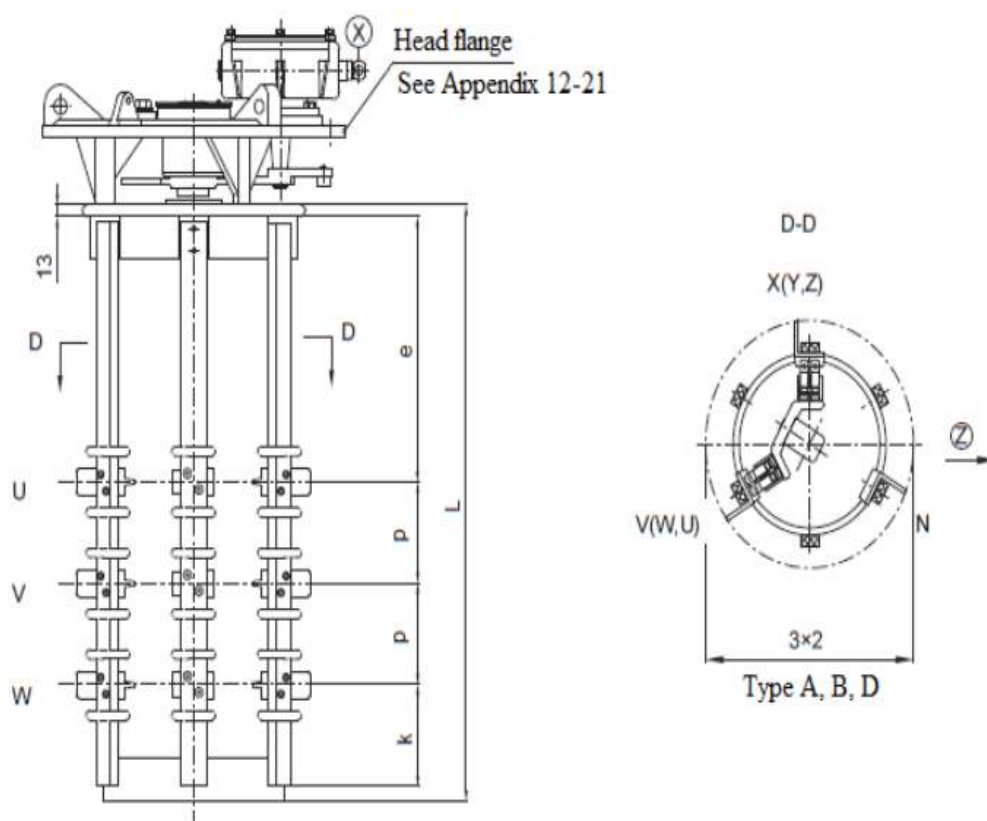
Appendix.3 600-1000A Single bridge jumper type tap-changer external dimensional drawing



Maximum equipment voltage	e	p	k	L
12 kV	200	130	135	630
72.5 kV	340	280	145	1080
126 kV	470	410	155	1480

Unit : mm

Appendix 4 600-1000A Y-D transform regulation, overall dimension



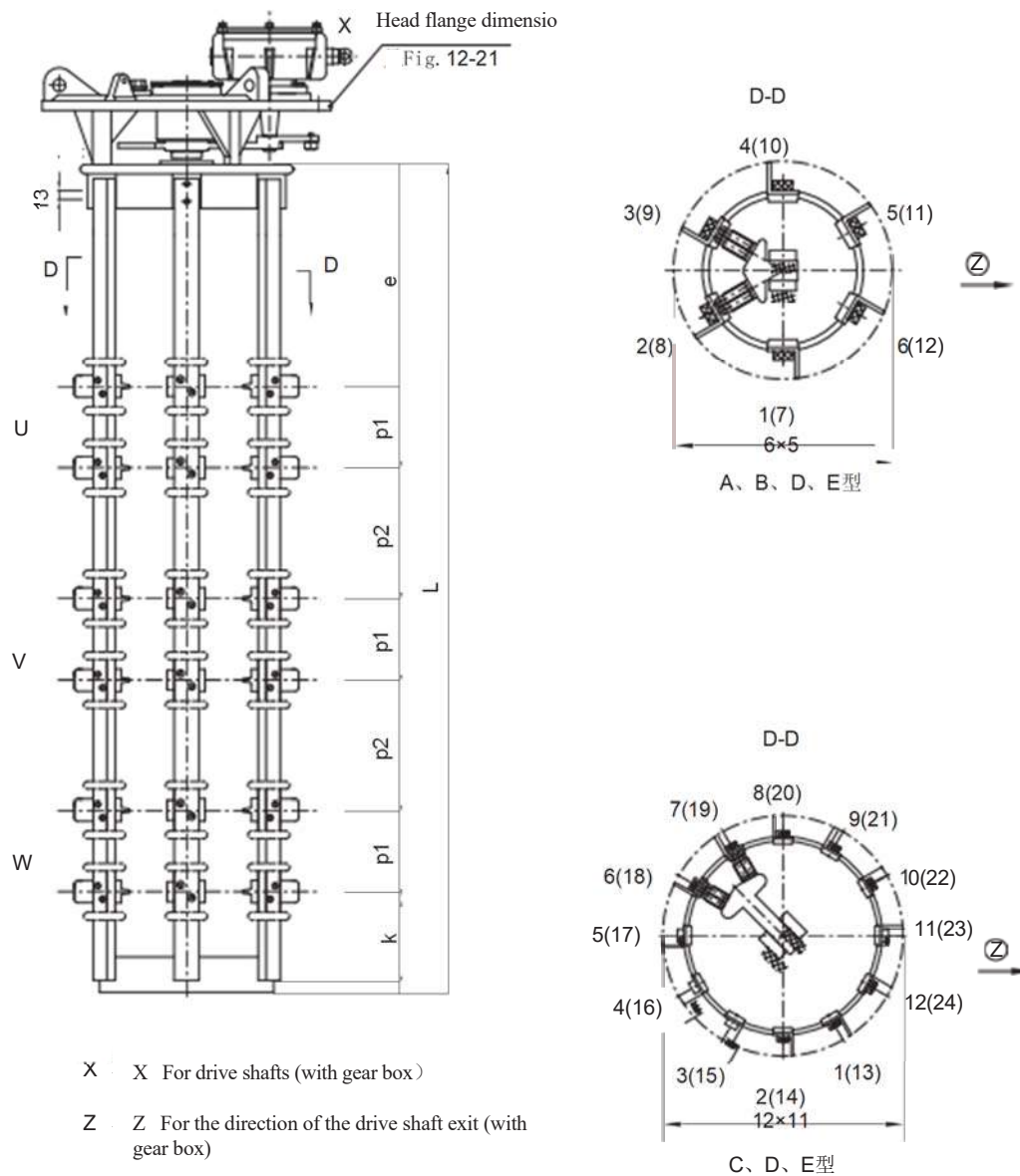
- (X) Driving shaft (with bevel gear)
 (Y) Direction of driving shaft (with bevel gear)

Note: ① Type A only for $I \leq 800A$
 ② Contact size see appendix 20
 ③ The size of A, B, D, E is $\Phi 350mm$, $\Phi 500mm$, $\Phi 600mm$ and $\Phi 750mm$

Highest voltage for equipment	e	p	k	L
12 kV	200	130	135	630
72.5 kV	340	280	145	1080
126 kV	470	410	155	1480

Unit: mm

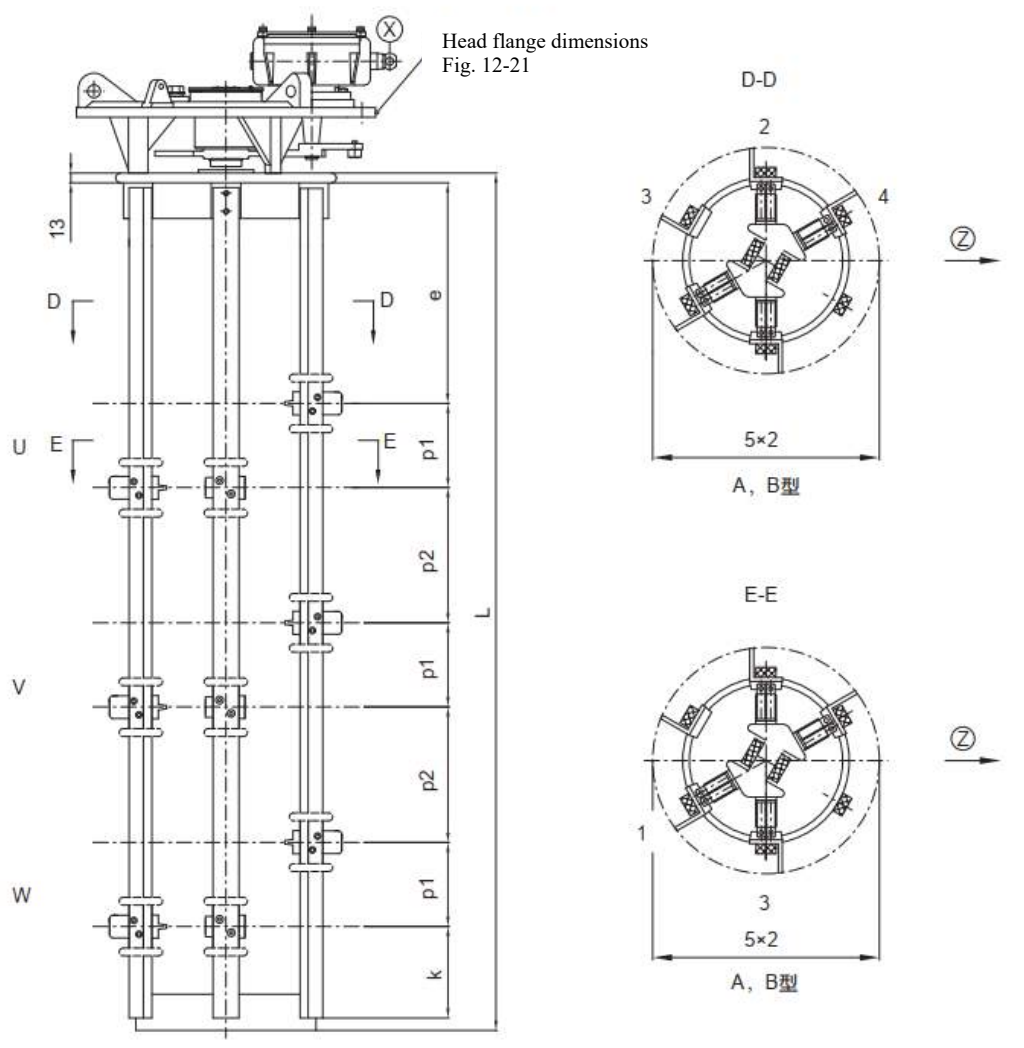
Appendix 5 600-1000A Double bridge jumper type tap-changer external dimensional drawing



Maximum equipment voltage	e	p1	p2	k	L
12 kV	200	120	150	125	1020
72.5 kV	340	160	280	145	1560
126 kV	470	170	410	155	1990

unit : mm

Appendix .6 600-1000A Dimensional drawing of the tap-changer with positive and negative adjustment



X For drive shafts (with gear box)

Z For the direction of the drive shaft exit (with gear box)

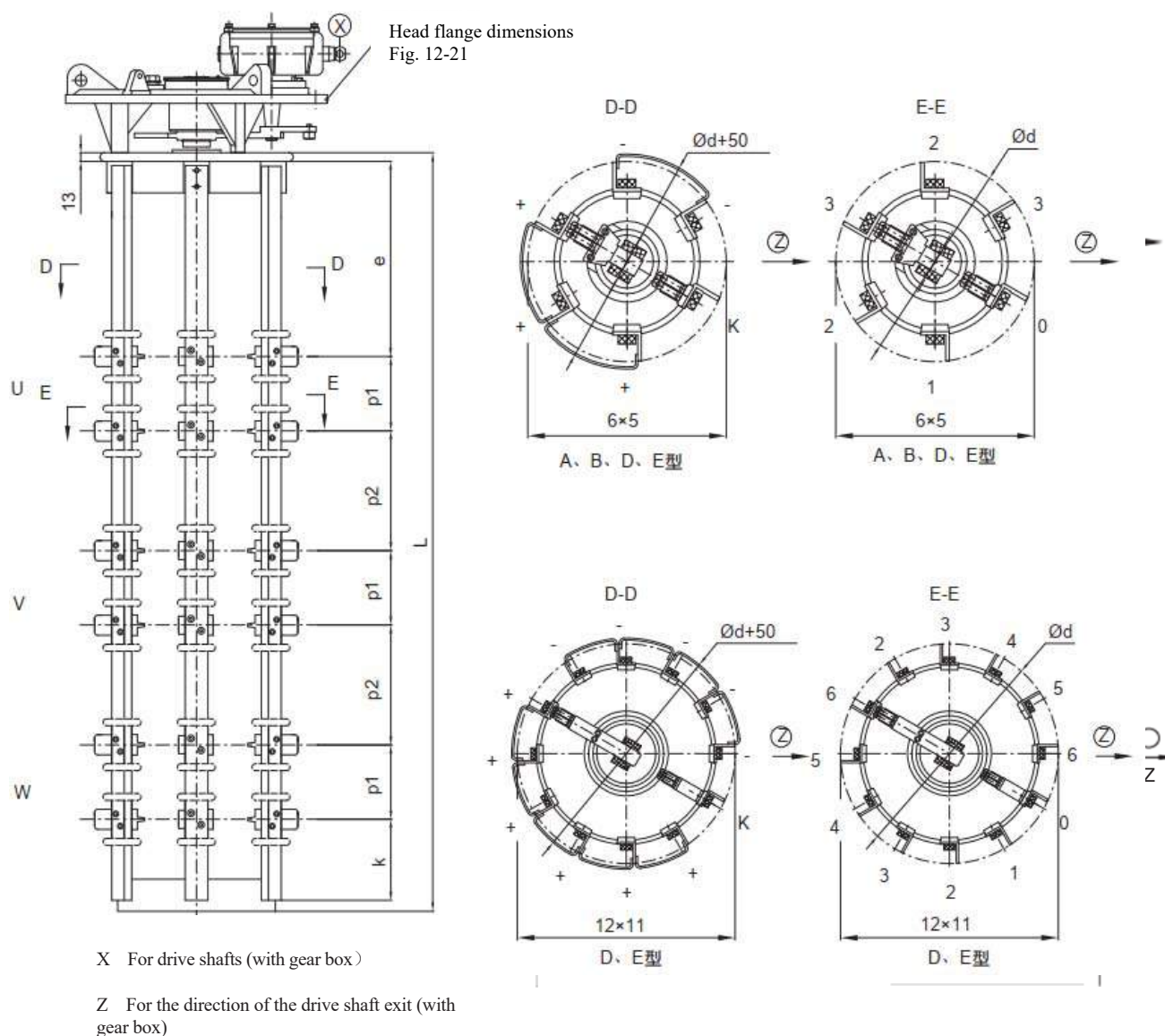
Note:

- 1.Type A only for currents ≤ 800 A
- 2.Contact size see Fig.20
- 3.The outer diameter of the cage contacts of types A, B, D and E are $\Phi 350\text{mm}$, $\Phi 500\text{mm}$, $\Phi 600\text{mm}$ and $\Phi 750\text{mm}$ respectively

Maximum equipment voltage	e	p1	p2	k	L
12 kV	200	120	150	125	1020
72.5 kV	340	160	280	145	1560
126 kV	470	170	410	155	1990

unit : mm

Appendix 7 600-1000A Dimensional drawing of the tap-changer with positive and negative adjustment

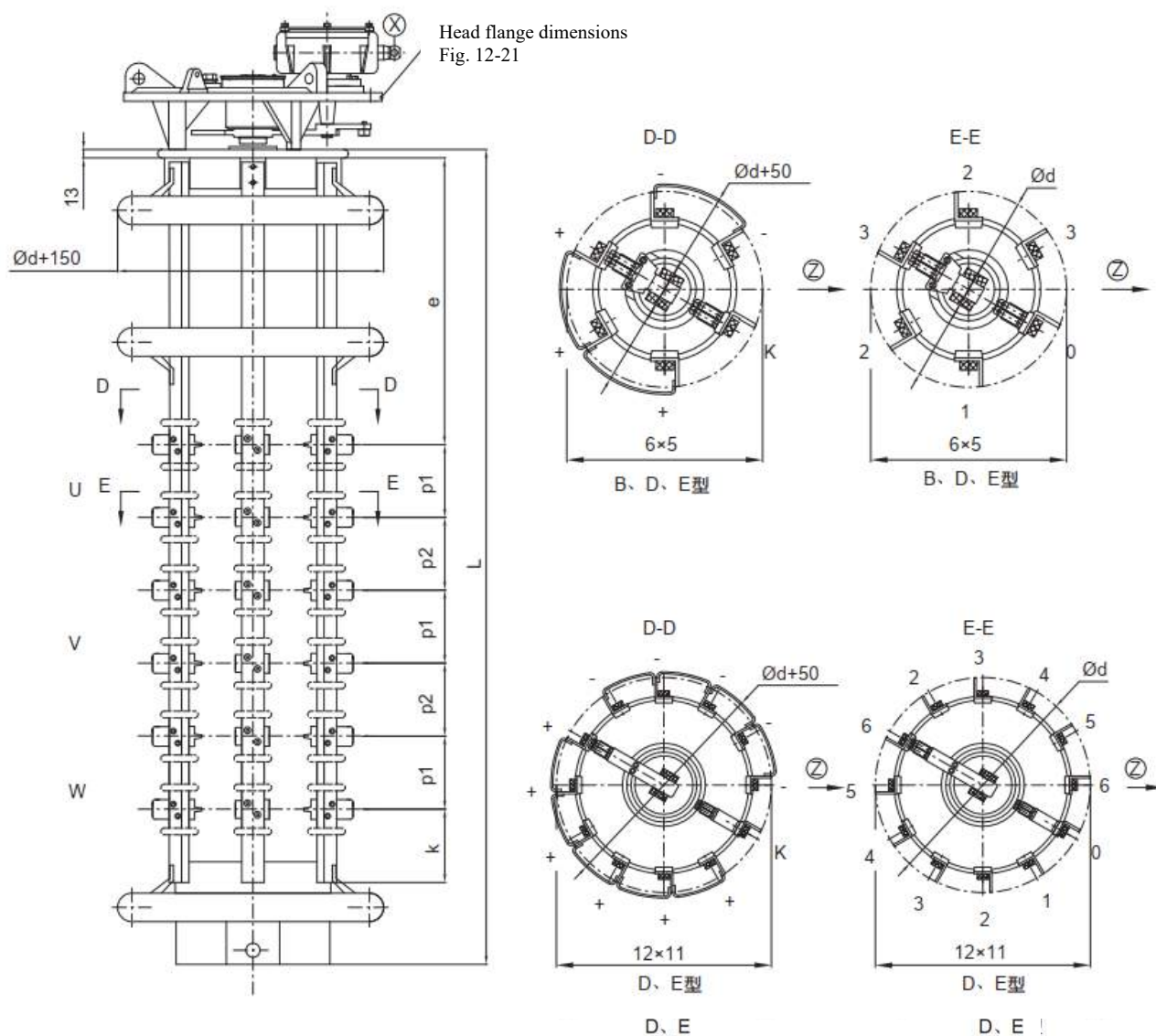


- Note
- 1.Type A only for currents ≤ 800 A
 - 2.Contact size see Fig.20
 - 3.Only "+" and "+", "-" and "-" are connected at the factory, as shown in the diagram "D-D", other users connect themselves
 - 4.The outer diameter of the cage contact is $\Phi 350$ mm, $\Phi 500$ mm, $\Phi 600$ mm and $\Phi 750$ mm for A, B, D and E types respectively.

Three-phase connection	WYE					Star				
	e	p1	p2	k	L	e	p1	p2	k	L
12 kV	170	120	120	125	930	200	120	150	125	1020
72.5 kV	340	135	160	150	1250	340	160	280	145	1560
126 kV	470	170	170	155	1510	470	170	410	155	1990

unit : mm

Appendix 8 600-1000A 252kV positive and negative tap changer dimensional drawing



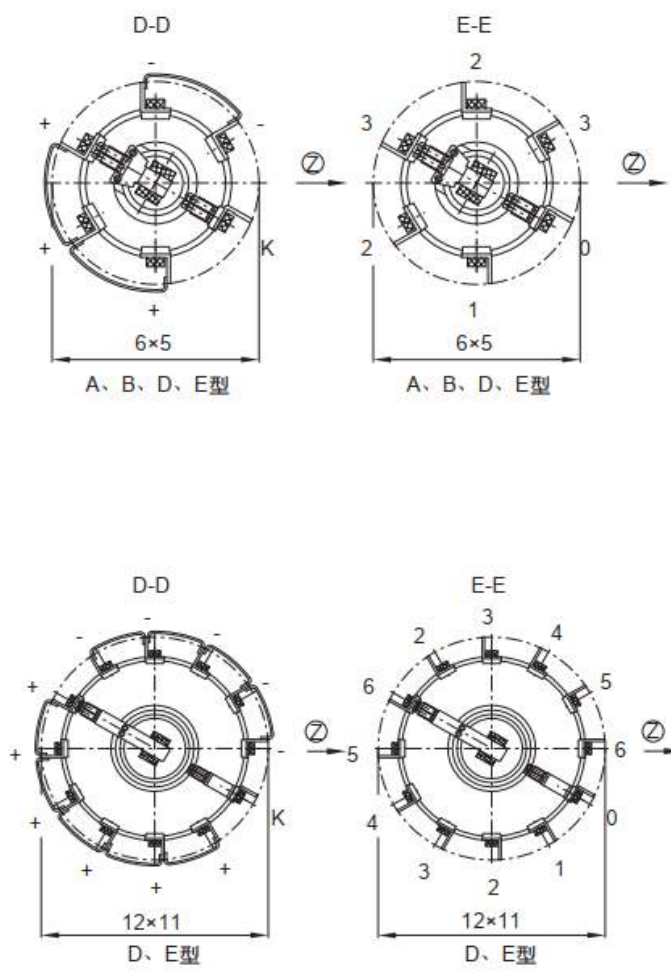
X For drive shafts (with gear box)

Z For the direction of the drive shaft exit (with gear box)

- Note
1. Type A only for currents ≤ 800 A
 2. Contact size see Fig.20
 3. Only "+" and "+", "-" and "-" are connected at the factory, as shown in the diagram "D-D", other users connect themselves
 4. The outer diameter of the cage contact is $\Phi 350\text{mm}$, $\Phi 500\text{mm}$, $\Phi 600\text{mm}$ and $\Phi 750\text{mm}$ for A, B, D and E types respectively.

Three-phase connection	WYE				
	e	p1	p2	k	L
Maximum equipment voltage					
252 kV	980	170	170	155	2200

unit :mm



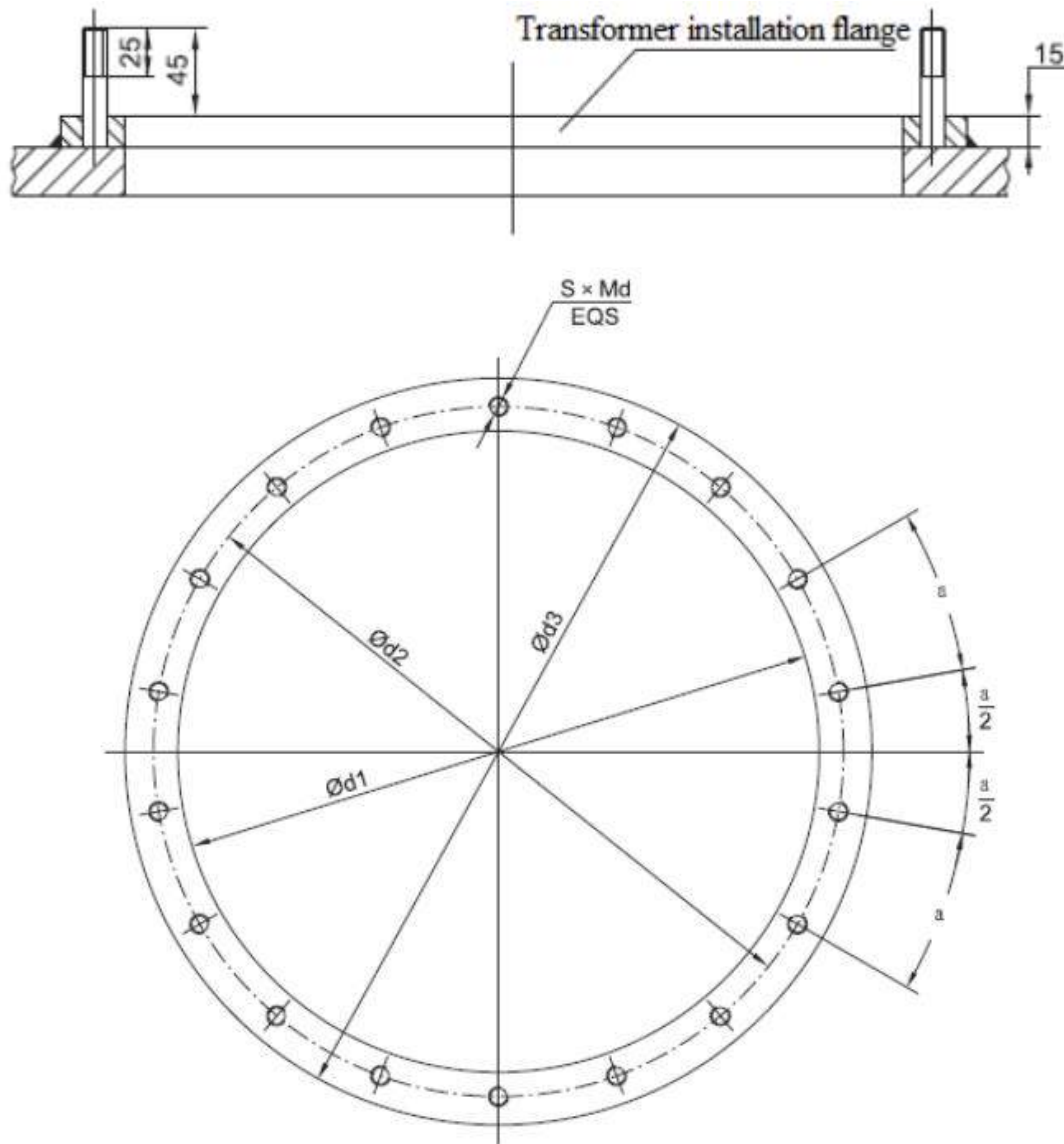
Z For the direction of the drive shaft exit (with gear box)

- 注: 1.Type A only for currents ≤ 800 A
2.Contact size see Fig.20
3.Only "+" and "+", "-" and "-" are connected at the factory, as shown in the diagram "D-D", other users connect themselves
4.The outer diameter of the cage contact is $\Phi 350\text{mm}$, $\Phi 500\text{mm}$, $\Phi 600\text{mm}$ and $\Phi 750\text{mm}$ for A, B, D and E types respectively.

Three-phase connection	WYE				
Maximum equipment voltage	e	p1	p2	k	L
12 kV	170	135	135	105	1525
72.5 kV	340	135	160	150	1790
126 kV	470	170	170	155	2050

44

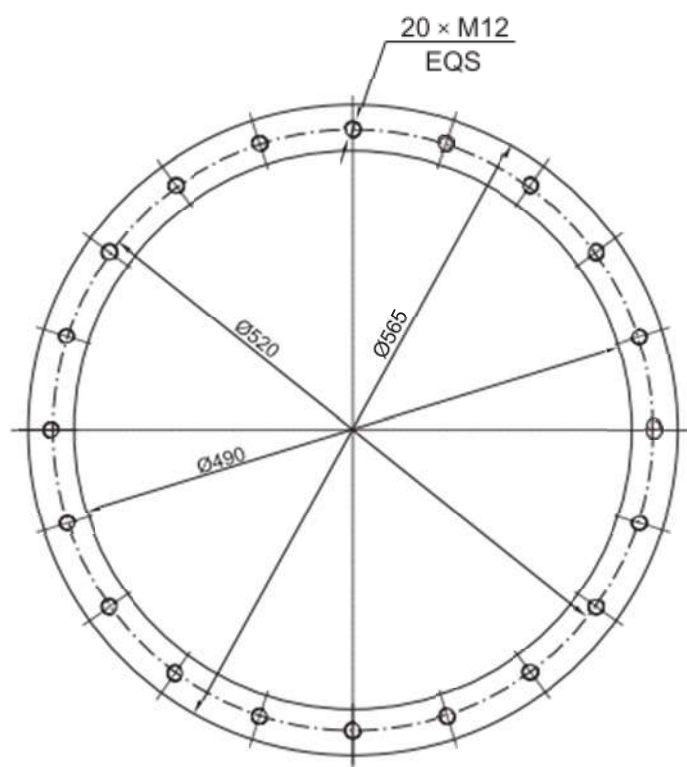
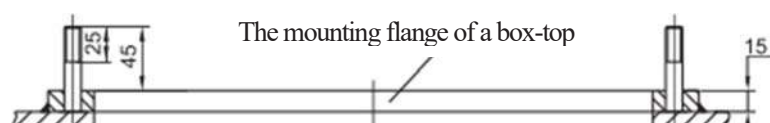
Appendix 10 Dimensional drawing of the mounting flange of a box-top transformer



Type	d1(Dia)	d2(Dia)	d3(Dia)	Screw distribution S-Md	Distribution angle a
A	Φ395	Φ425	Φ460	18×M12	20°
B	Φ520	Φ550	Φ590	20×M12	18°
D	Φ620	Φ650	Φ690	20×M12	18°
E	Φ770	Φ800	Φ840	20×M12	18°

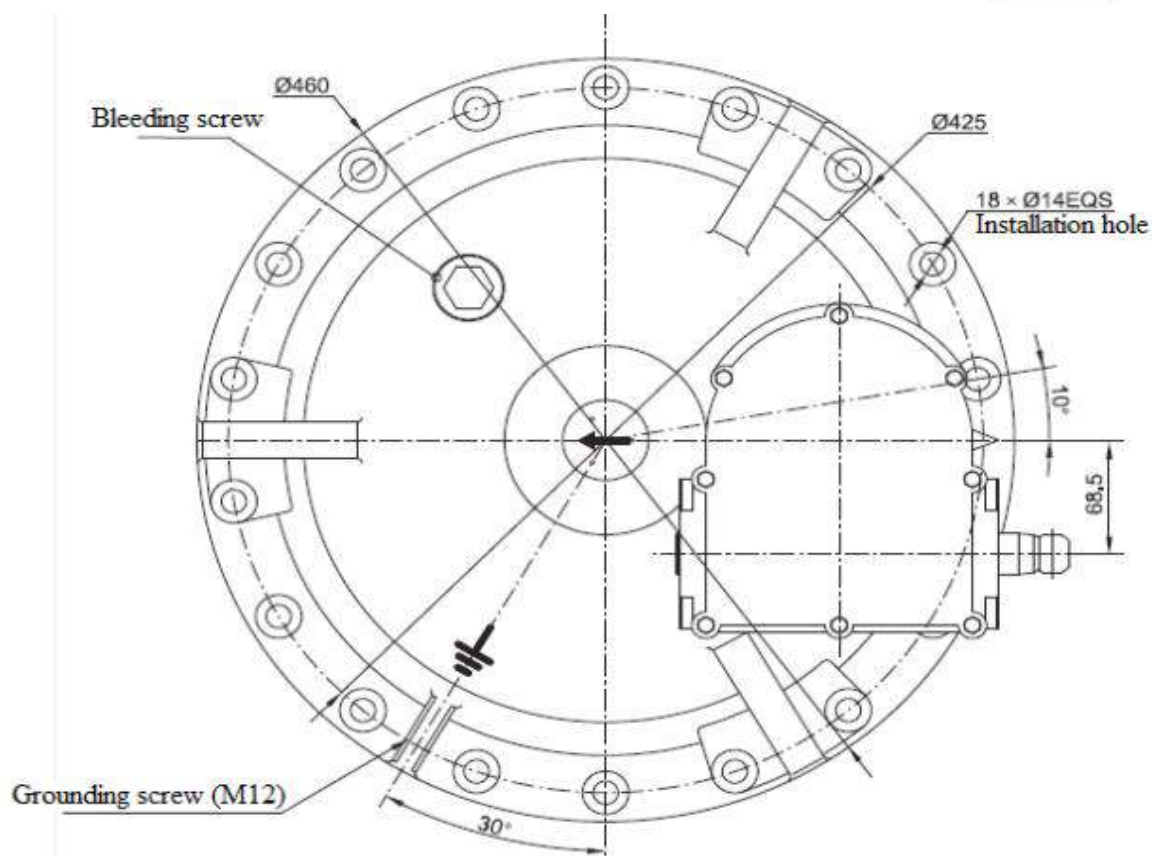
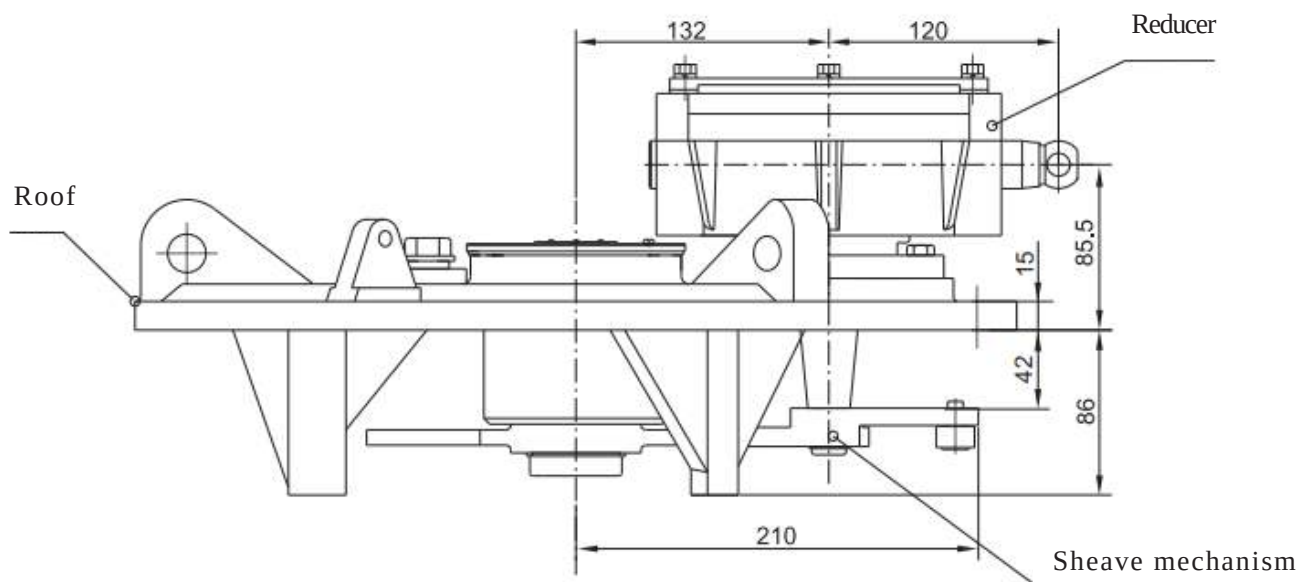
Unit: mm

Appendix 11 Bell-type transformer mounting flange dimensions



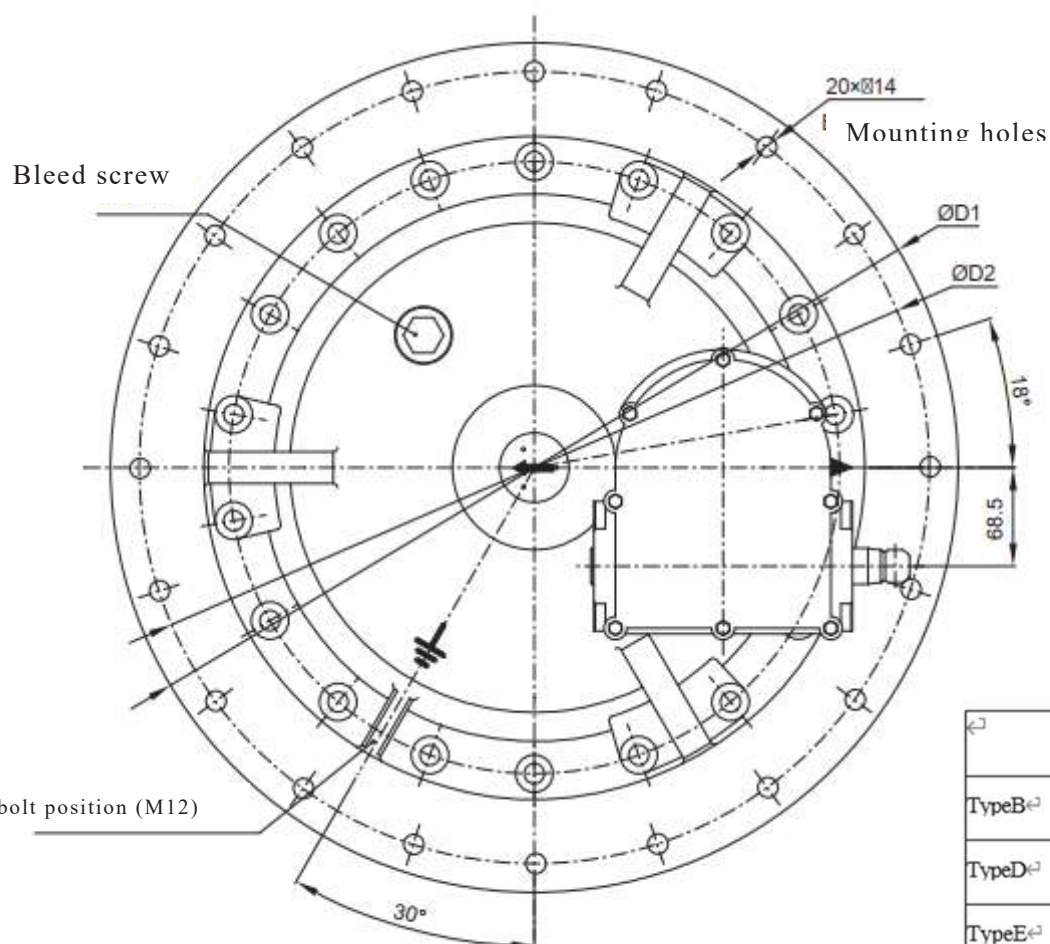
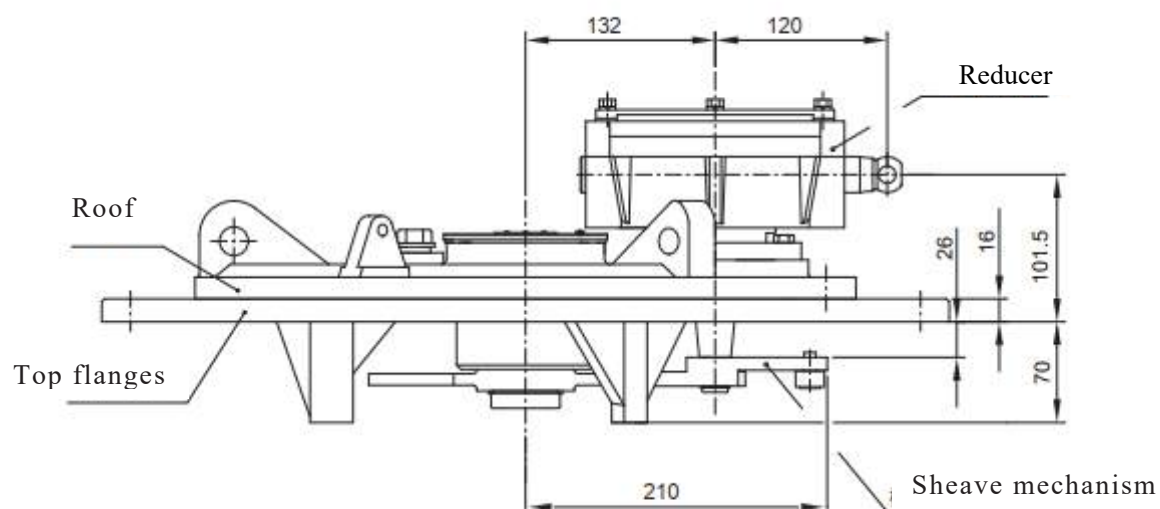
unit :mm

Appendix 12 Floor mounted motorized (manual) tap-changer (type A roof-mounted) head flange mounting dimensions



Unit: mm

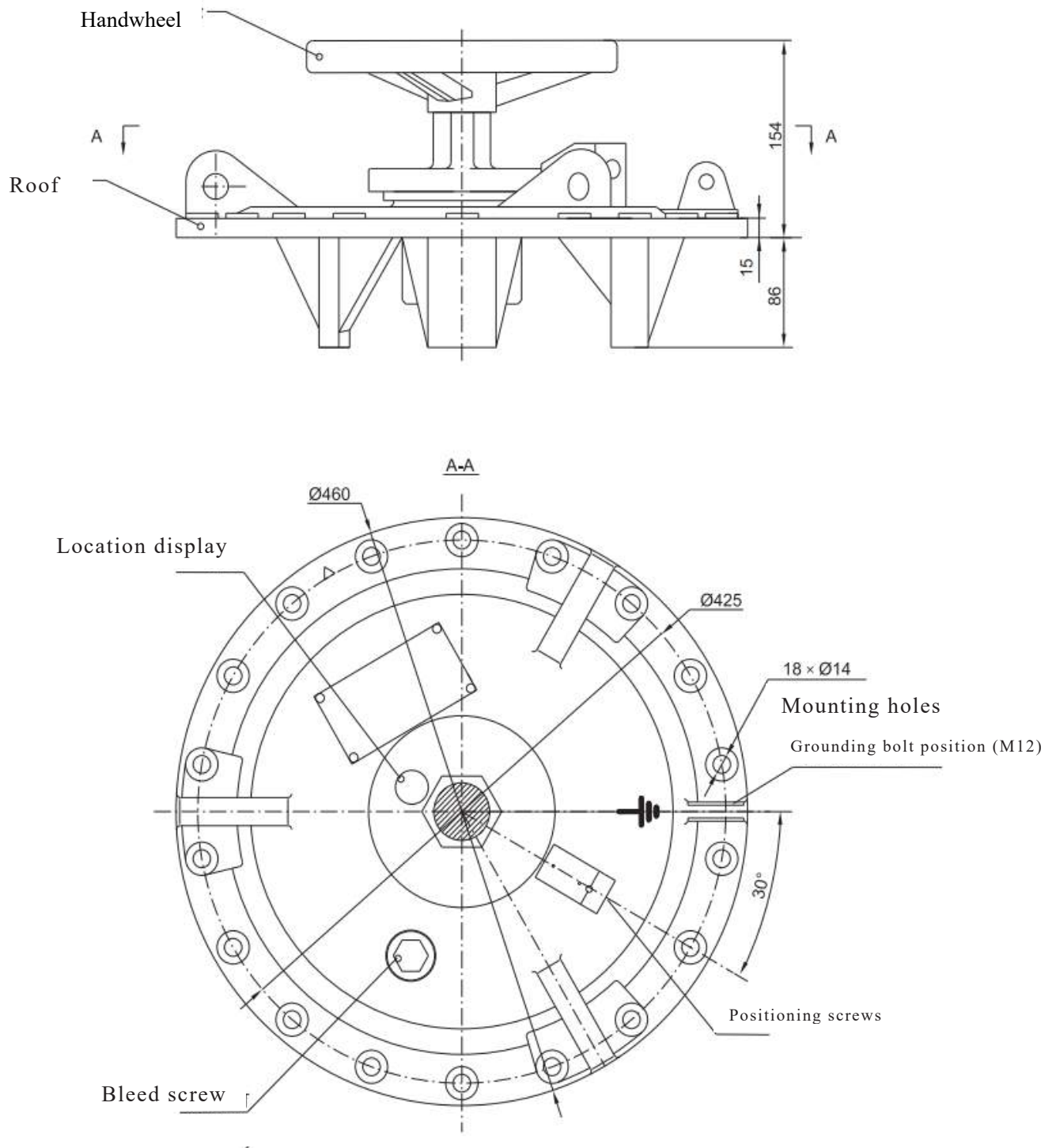
Appendix 13 Floor mounted motorized (manual) tap-changer (box top type B, D, E) head flange mounting dimensions



	ØD1	ØD2
TypeB	590	550
TypeD	690	650
TypeE	840	800

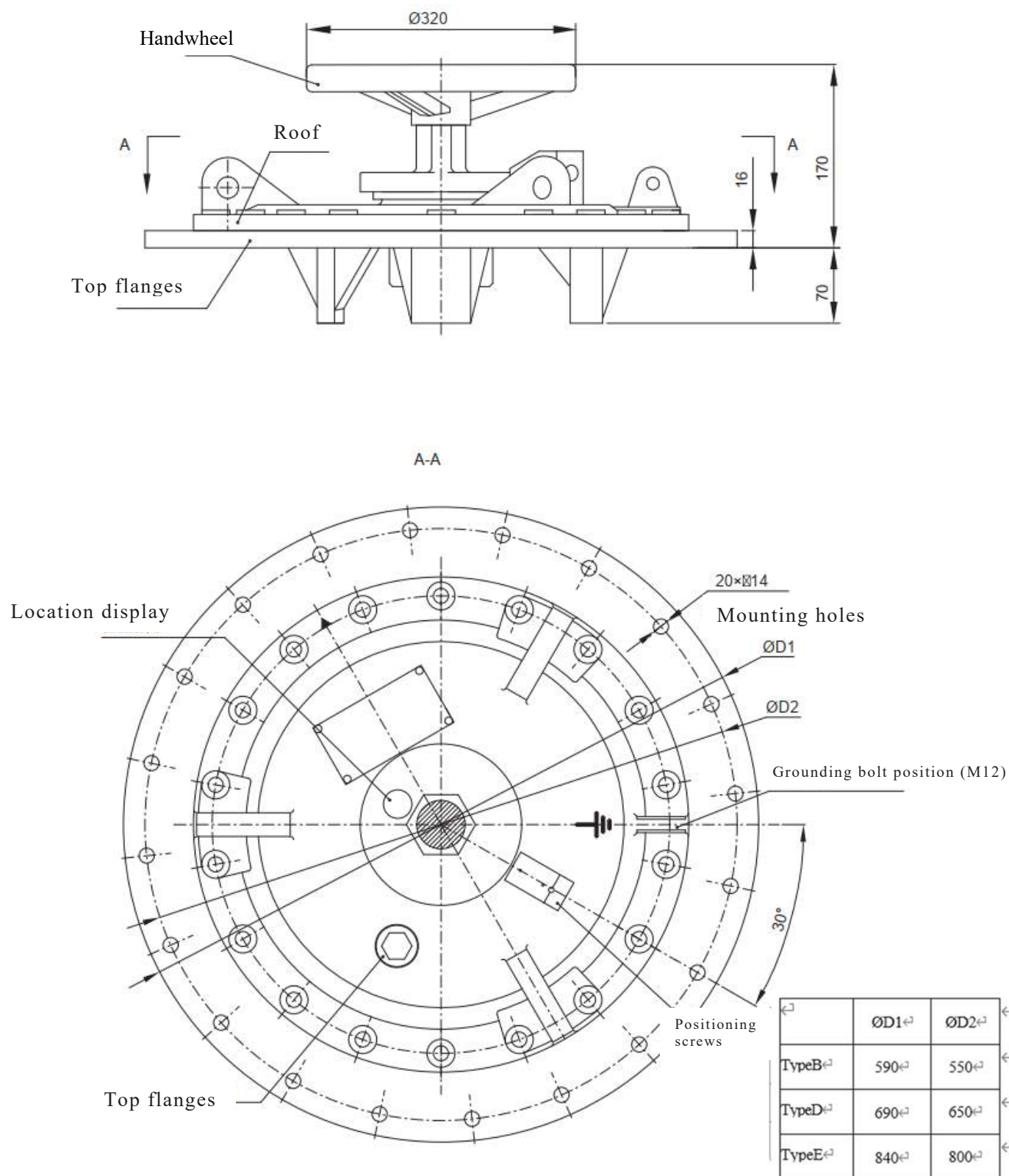
unit : mm

Appendix 14 Head flange mounting dimensions of the top cover handwheel tap changer (type A box top type)

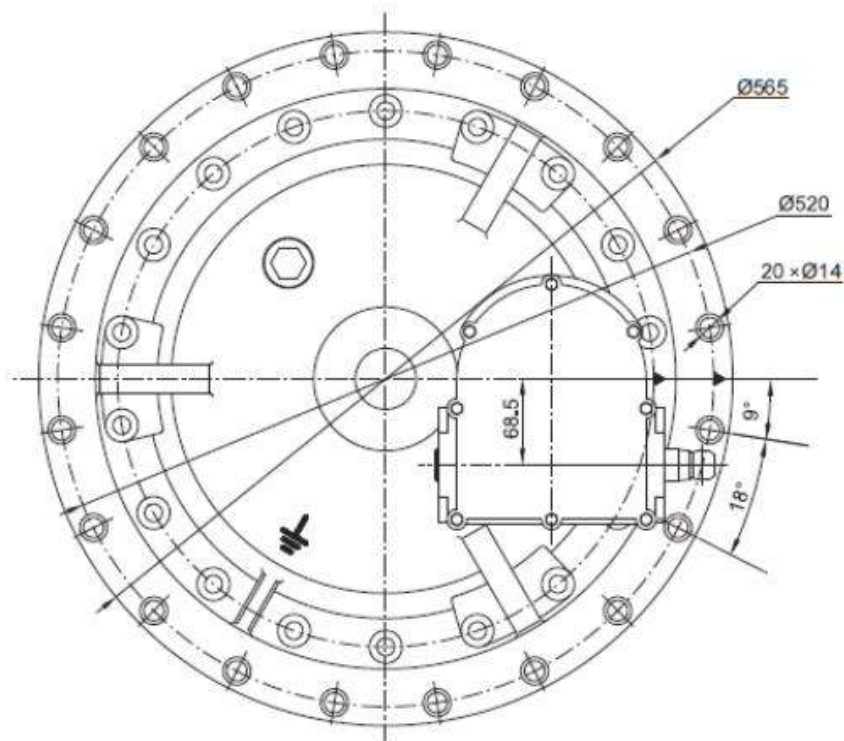
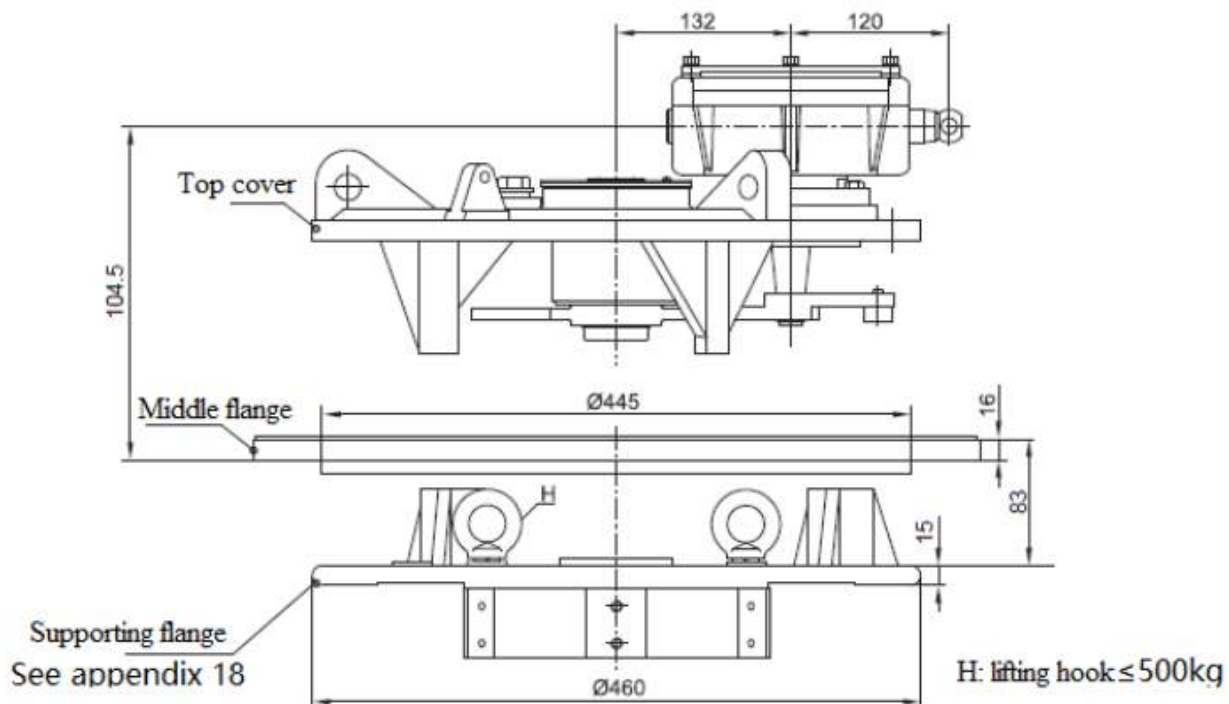


unit : mm

Appendix 15 Head flange mounting dimensions for top cover handwheel tap changer (type B, D, E box top)

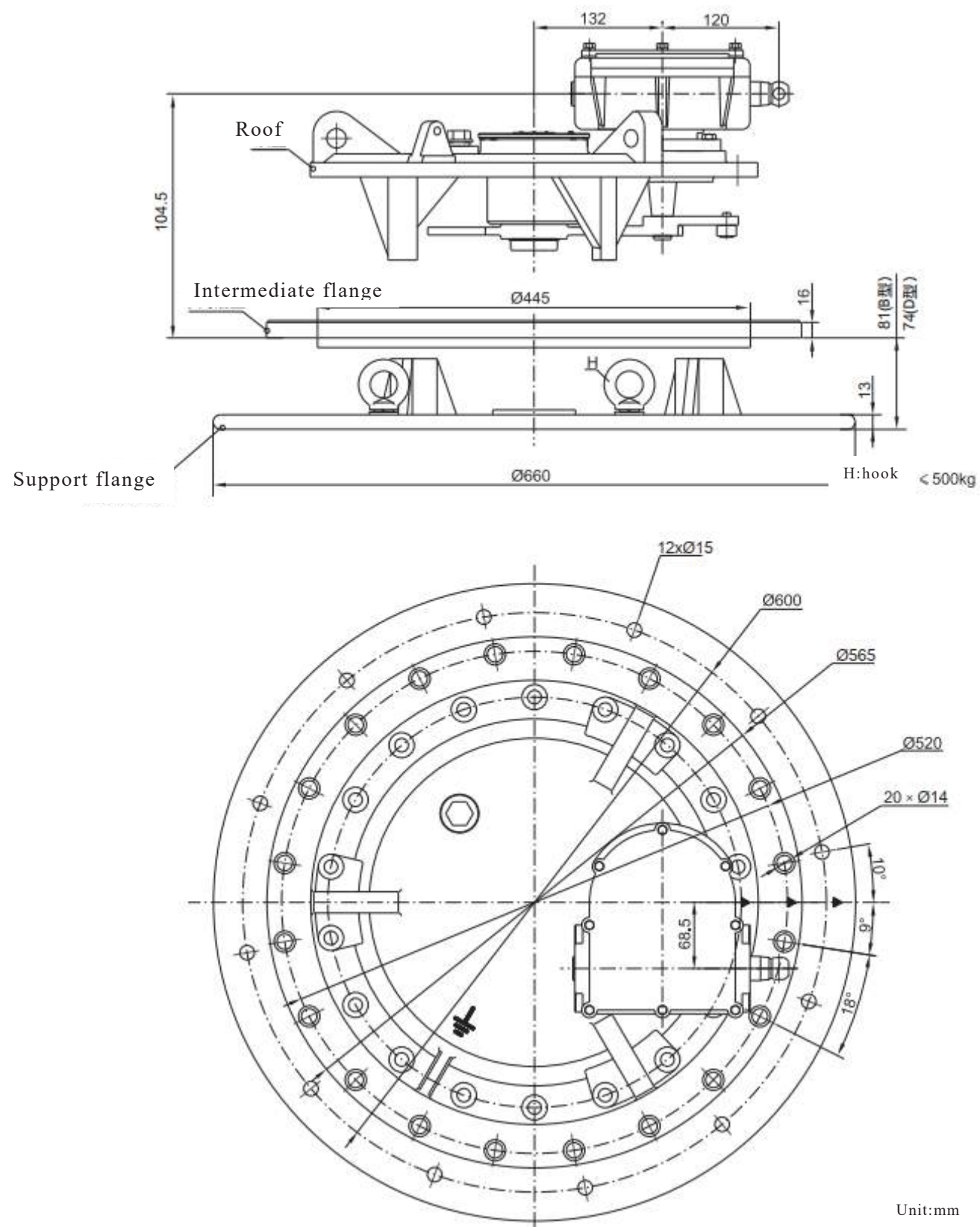


Appendix 16 Floor mounted motorized (manual) tap-changer (bell-type A) head flange mounting dimensions

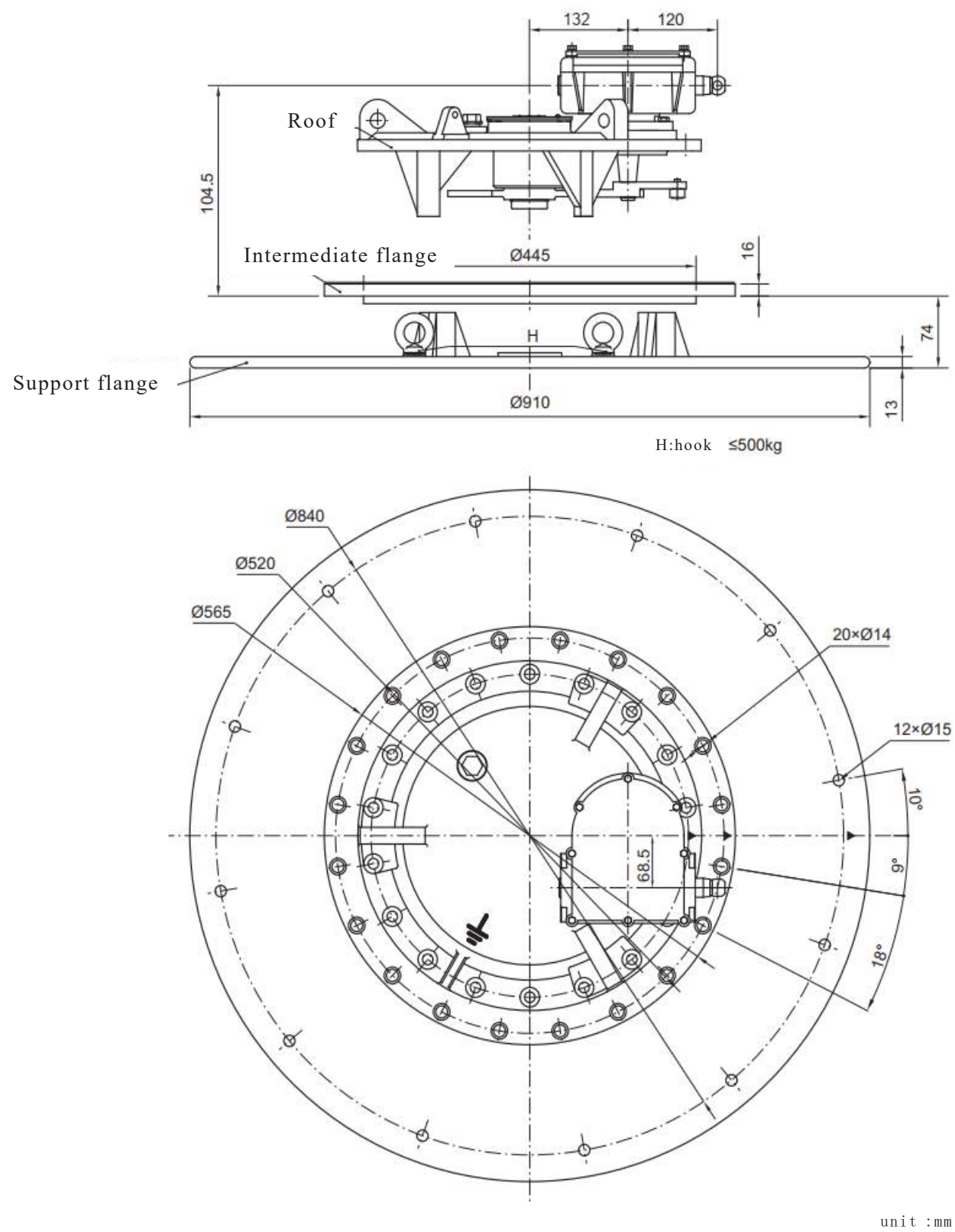


Unit: mm

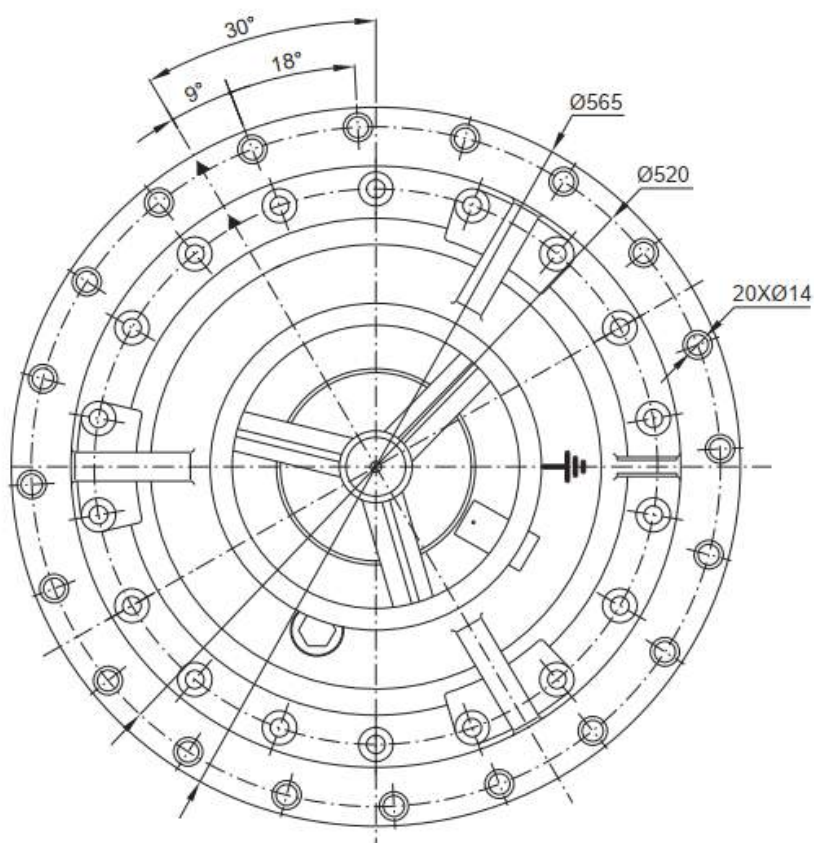
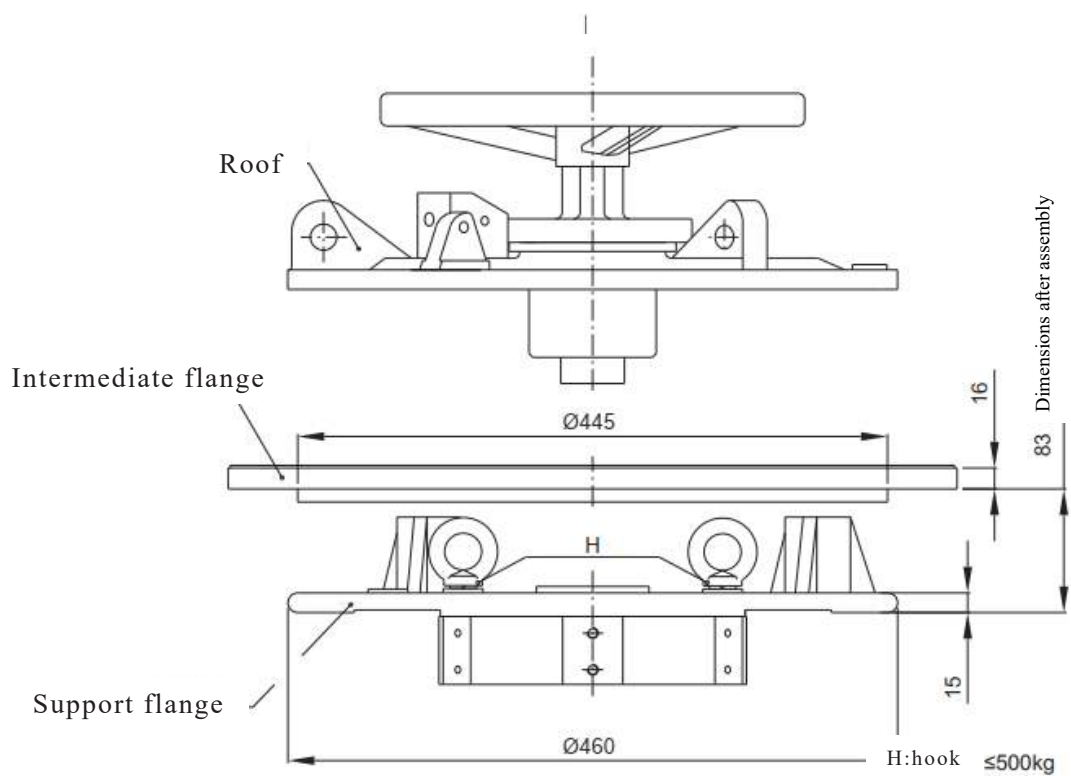
Appendix 17 Floor mounted motorized (manual) tap-changer (bell-type B, D) head flange mounting dimensions



Appendix 18 Floor mounted motorized (manual) tap-changer (bell-type E) head flange mounting dimensions

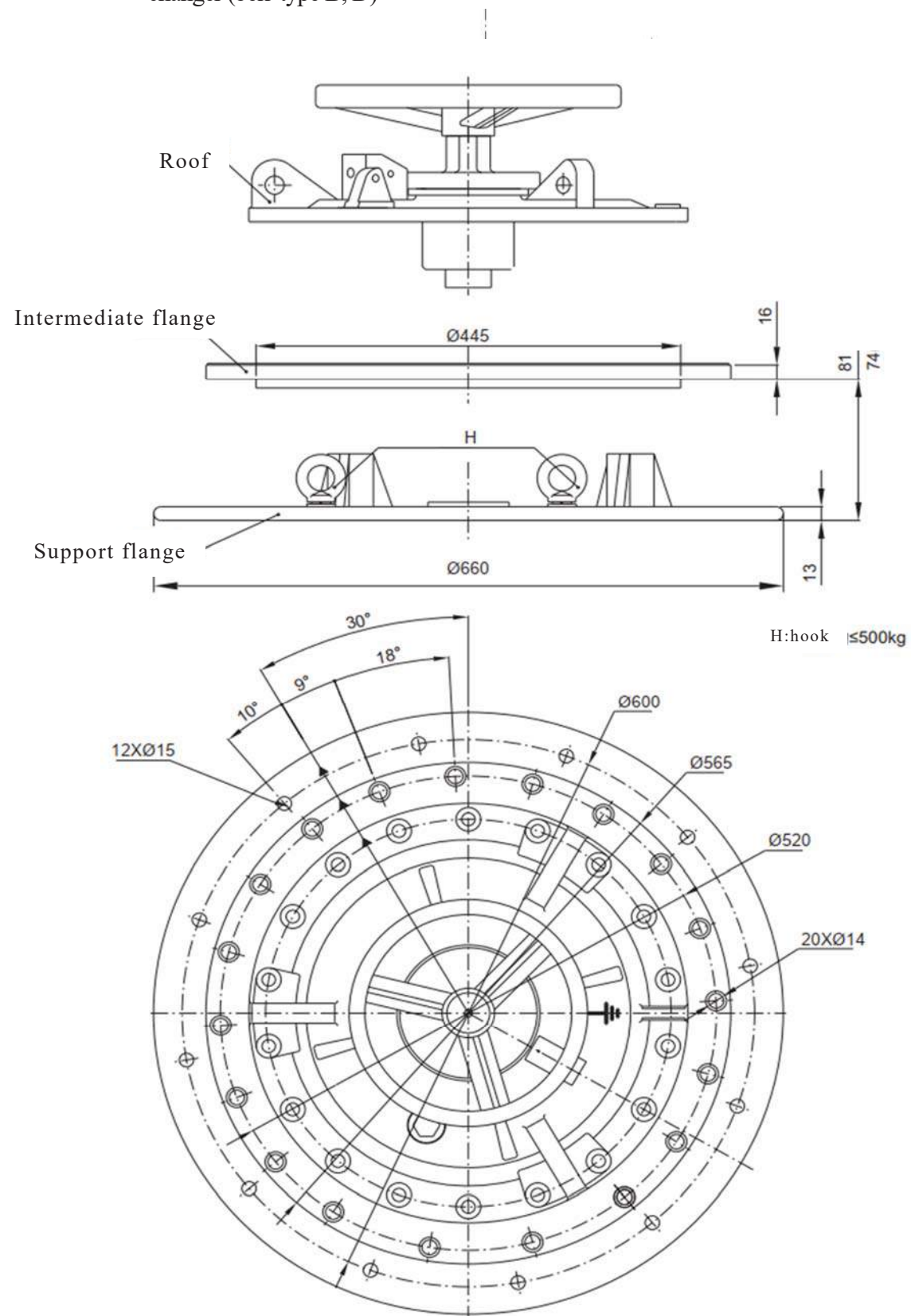


Appendix 19 Head flange mounting dimensions of the top cover handwheel tap changer (bell-type A)



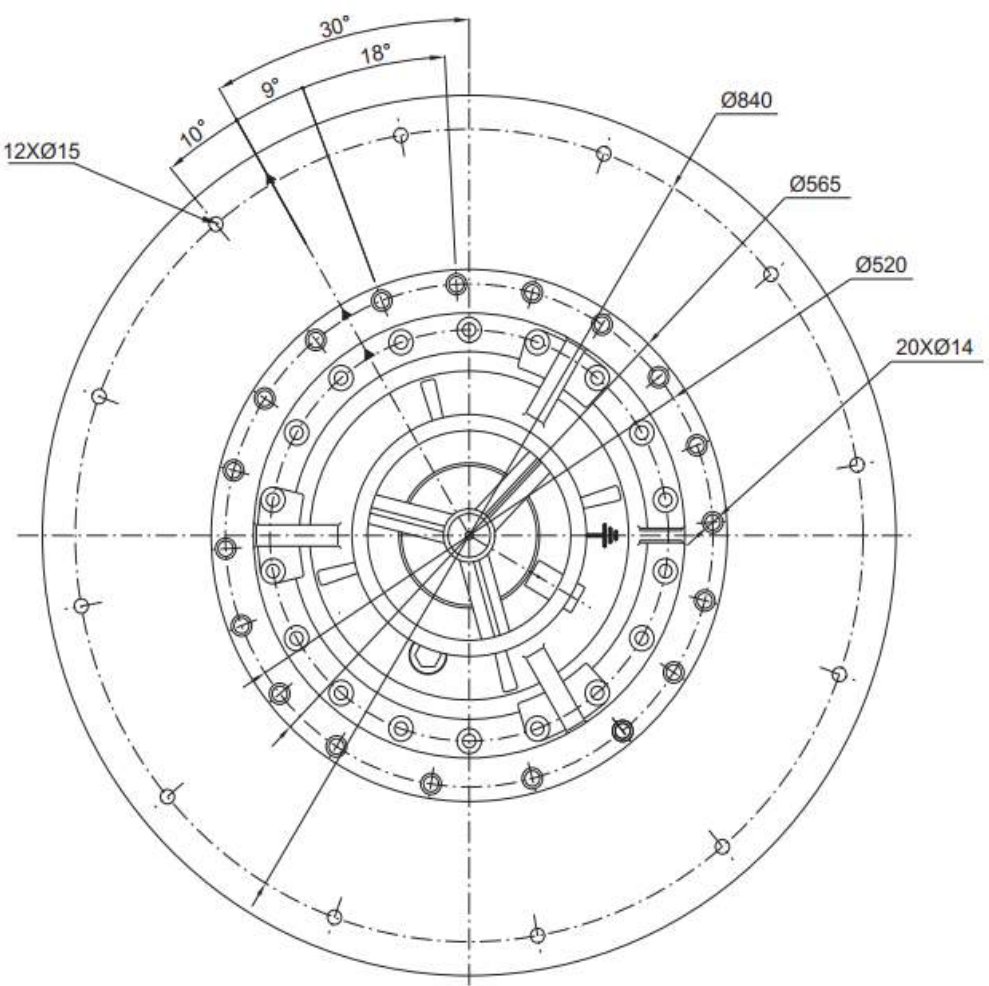
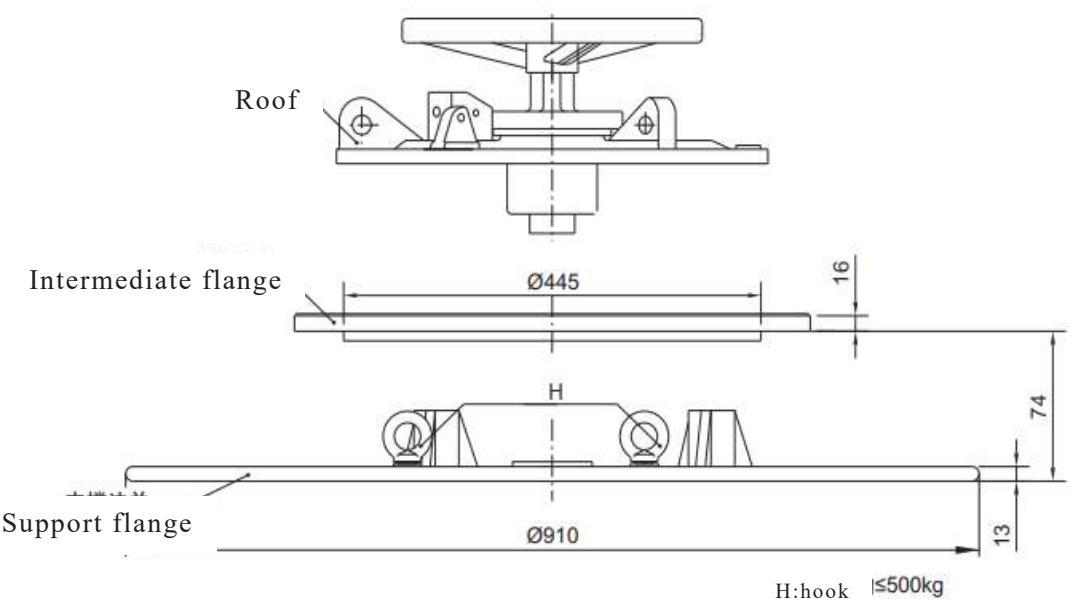
unit : mm

Appendix 20 Head flange mounting dimensions of the top cover handwheel tap changer (bell-type B, D)



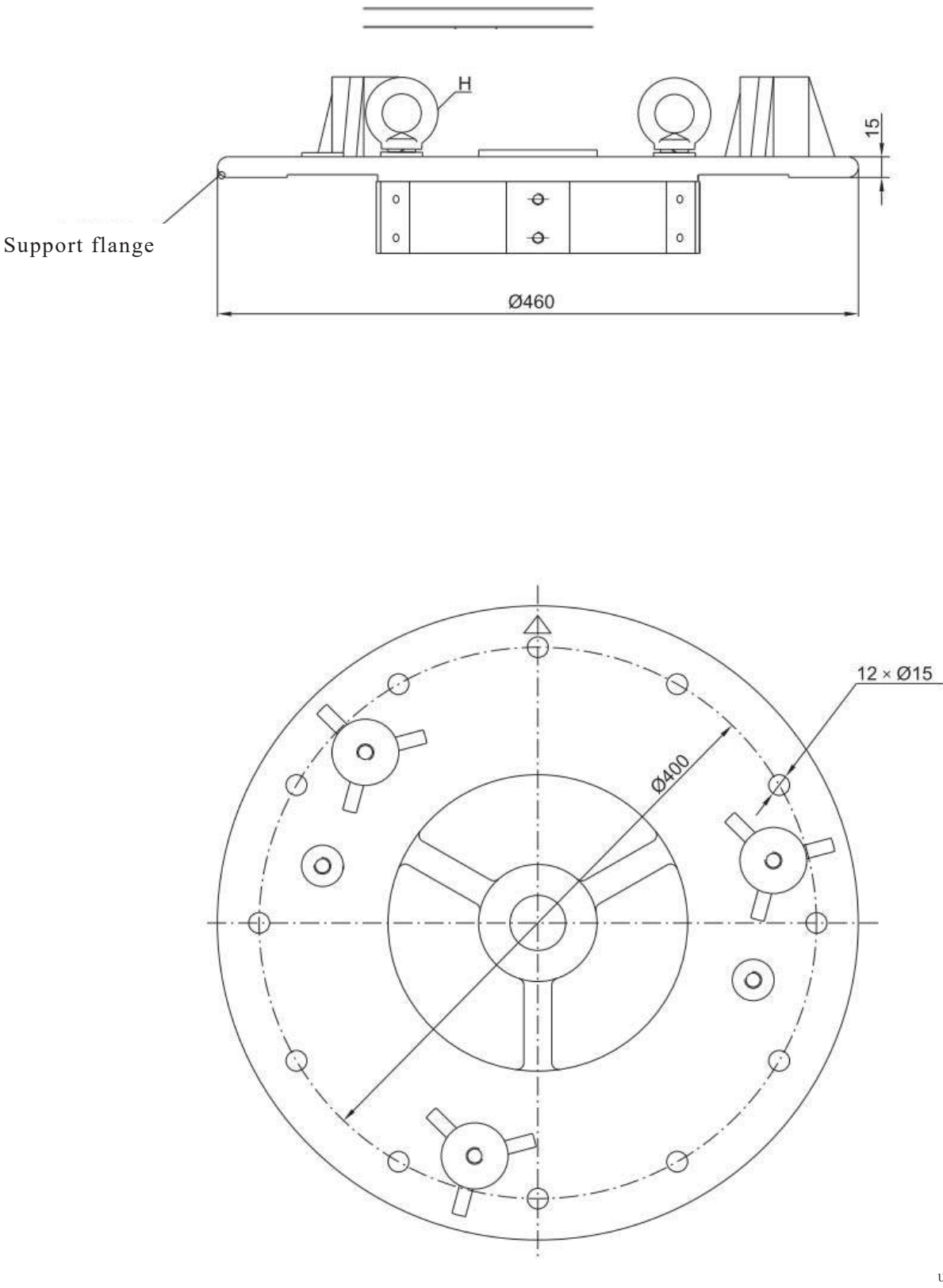
Unit:mm

Appendix 21 Head flange mounting dimensions of the top cover handwheel tap changer (bell-type E)

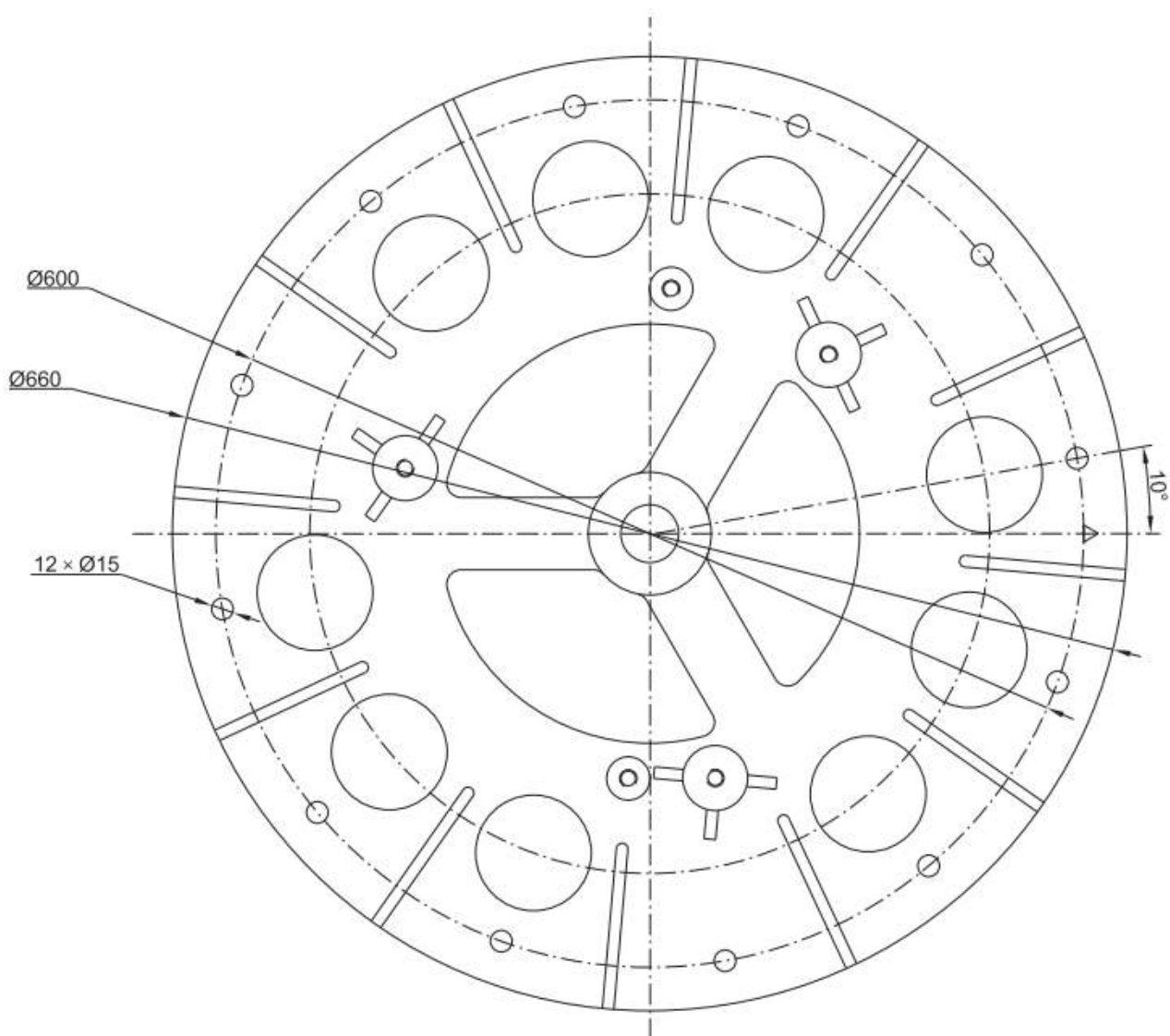
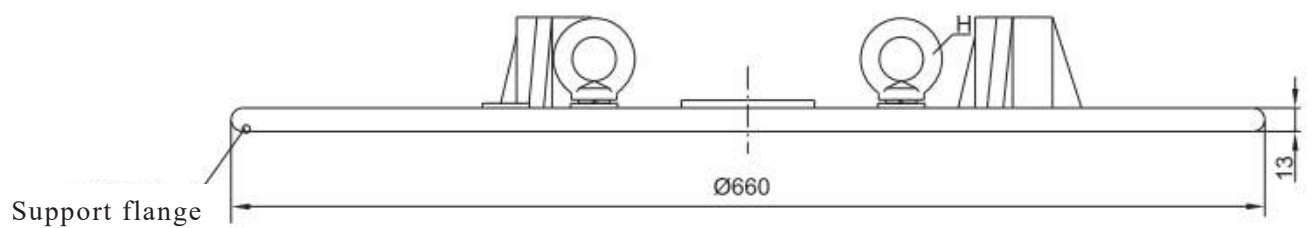


unit :mm

Appendix 22 Dimensional drawing of the supporting flange of the bell-type A tap-changer

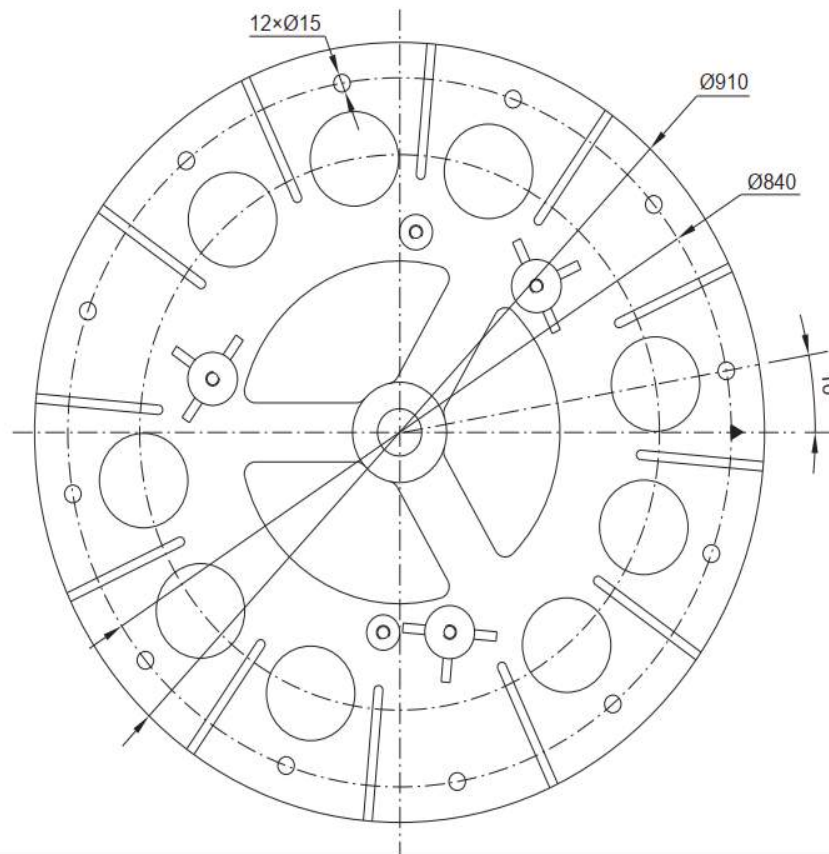
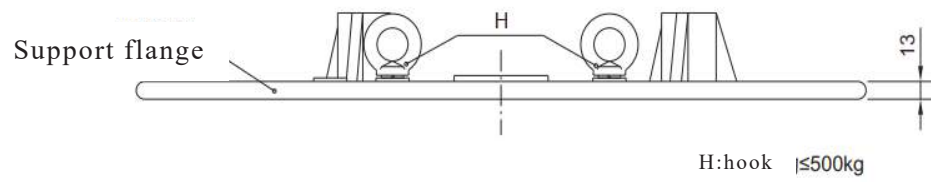


Appendix 23 Dimensional drawing of the supporting flange of the bell-type tap-changer type B,D



unit : mm

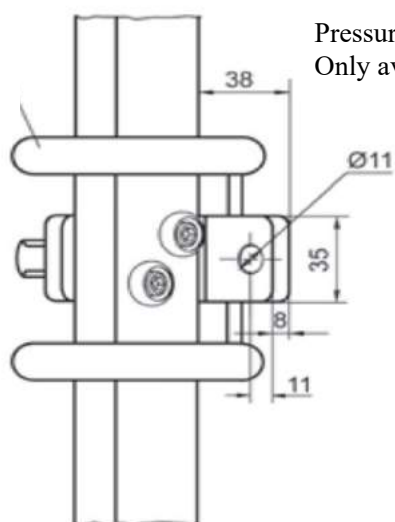
Appendix 24 Dimensional drawing of the supporting flange of the bell-type E tap-changer



Unit:mm

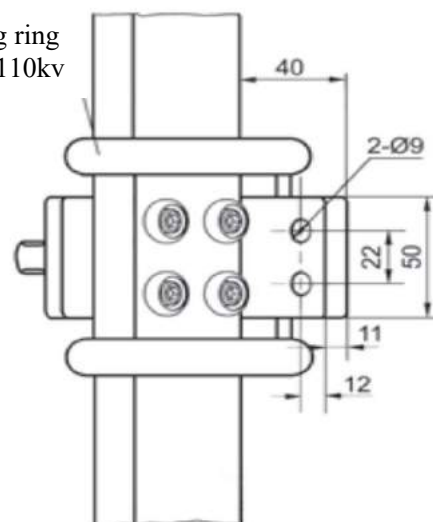
Appendix 25 Dimensional drawing of the tap changer terminals

Pressure equalizing ring
Only available on 110kv



300A/600A

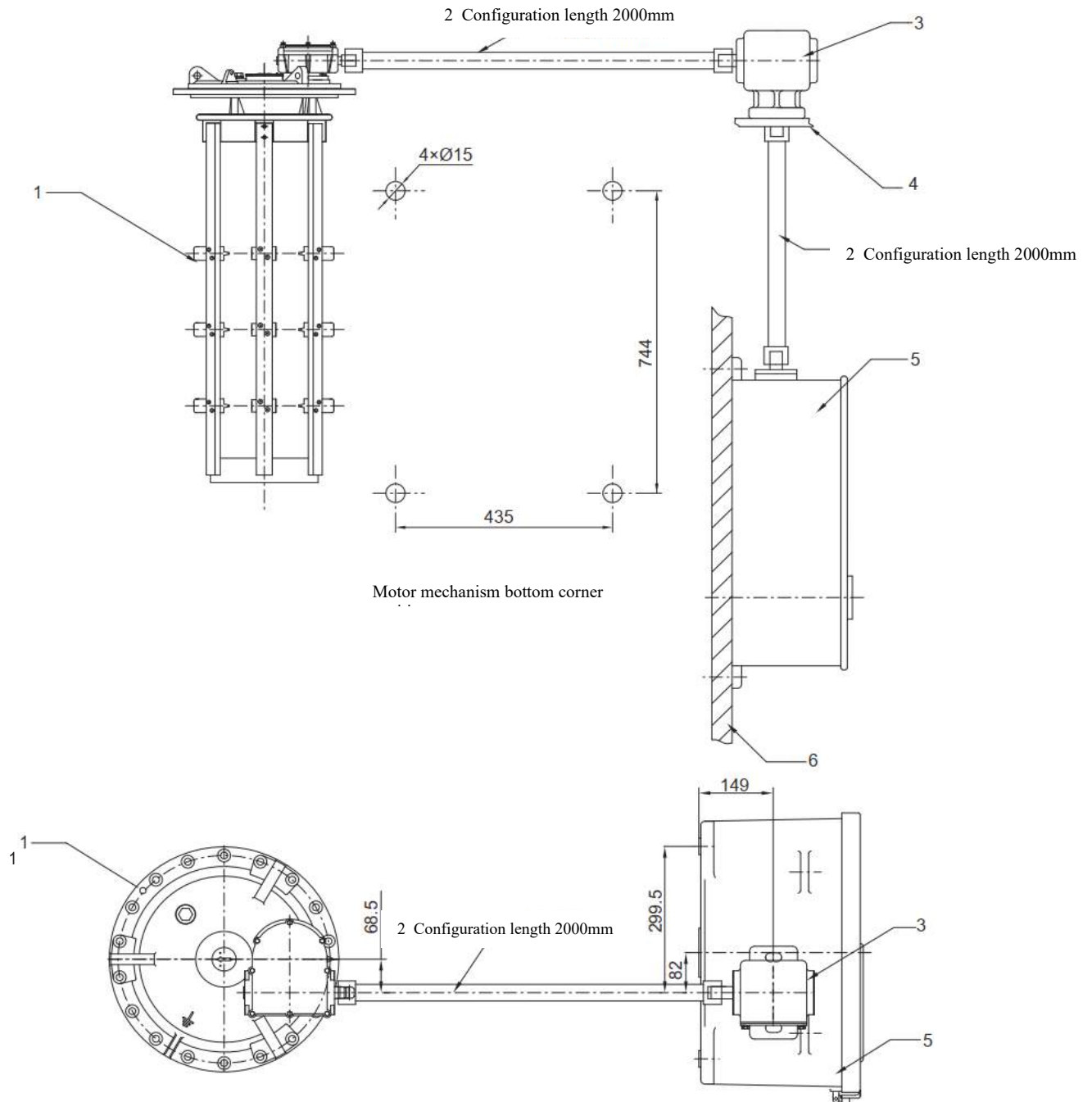
Pressure equalizing ring
Only available on 110kv



800A/1000A

unit : mm

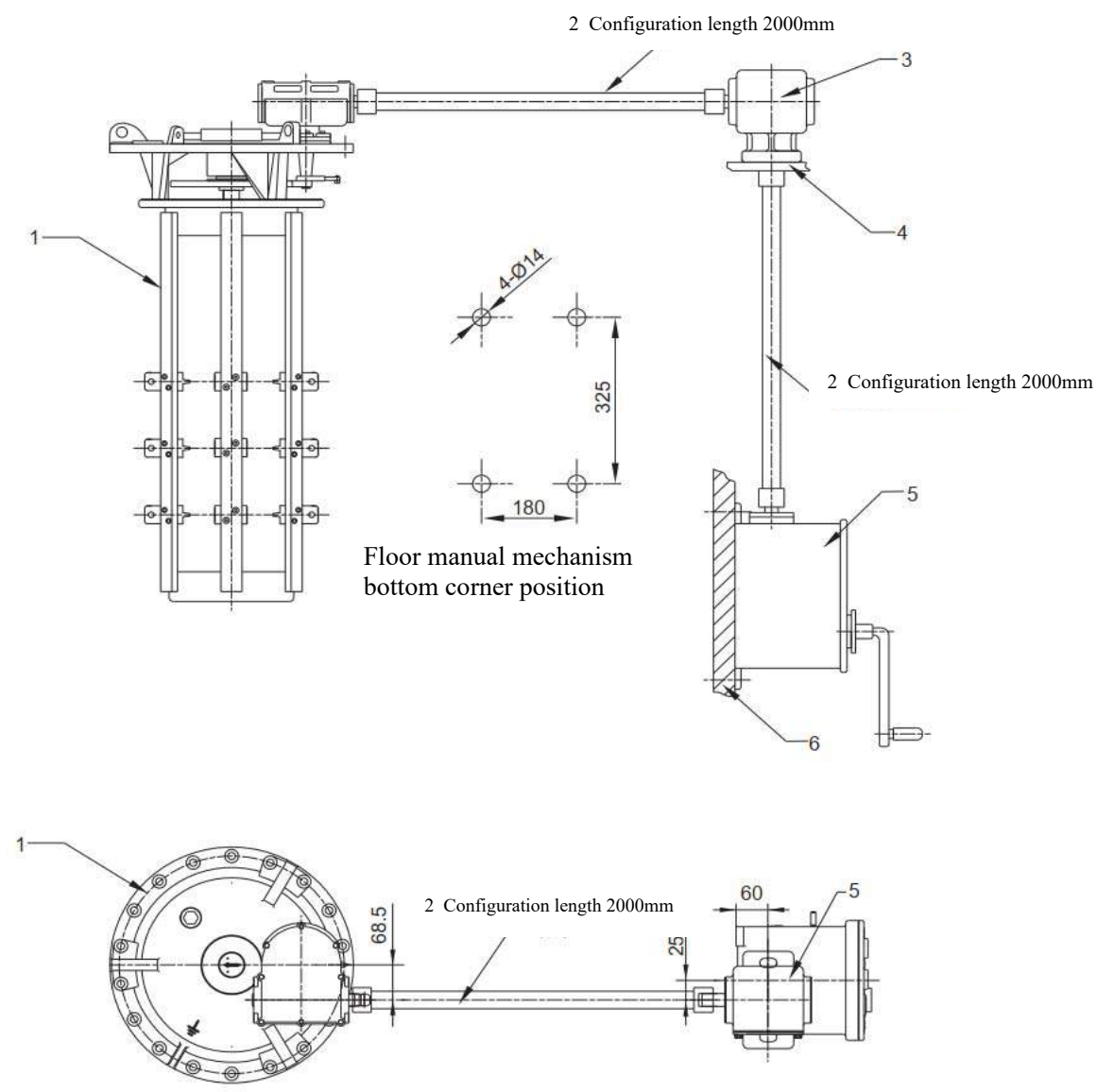
Appendix 26 Installation diagram for floor mounted motorized tap-changer



1. Switch body	2. Drive shafts	3. Bevel gearboxes
4. Mounting support plate (user supplied)	5. SHM-D motor mechanism (with SHM-K remote controller)	6. Mounting support plate (user supplied)

unit :mm

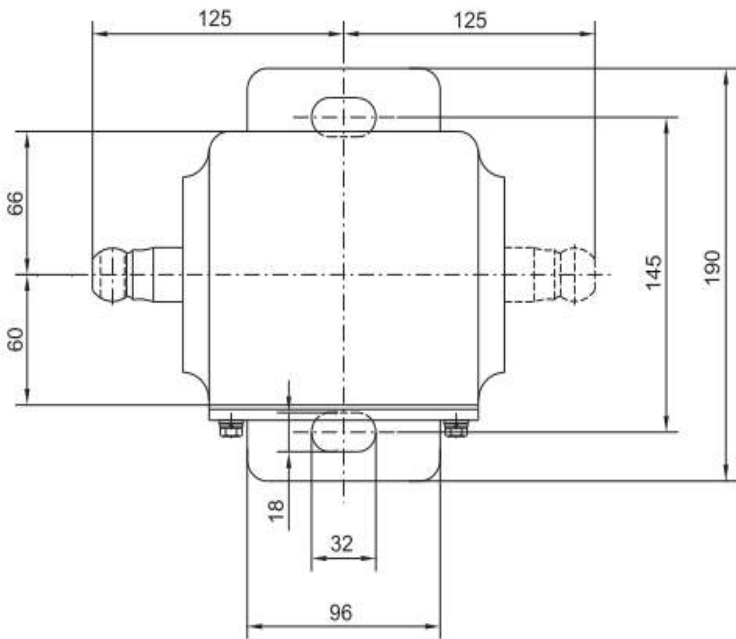
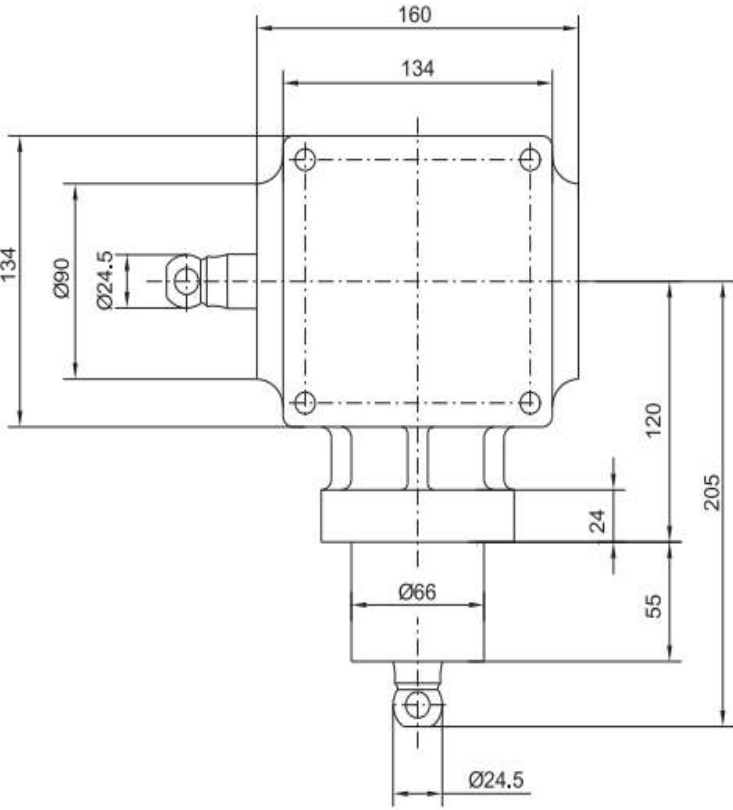
Appendix 27 Installation diagram for floor mounted manual tap-changer



1. Switch body	2. Drive shafts	3. Bevel gearboxes
4. Mounting support plate (user supplied)	5. SL agency	6. Mounting support plate (user supplied)

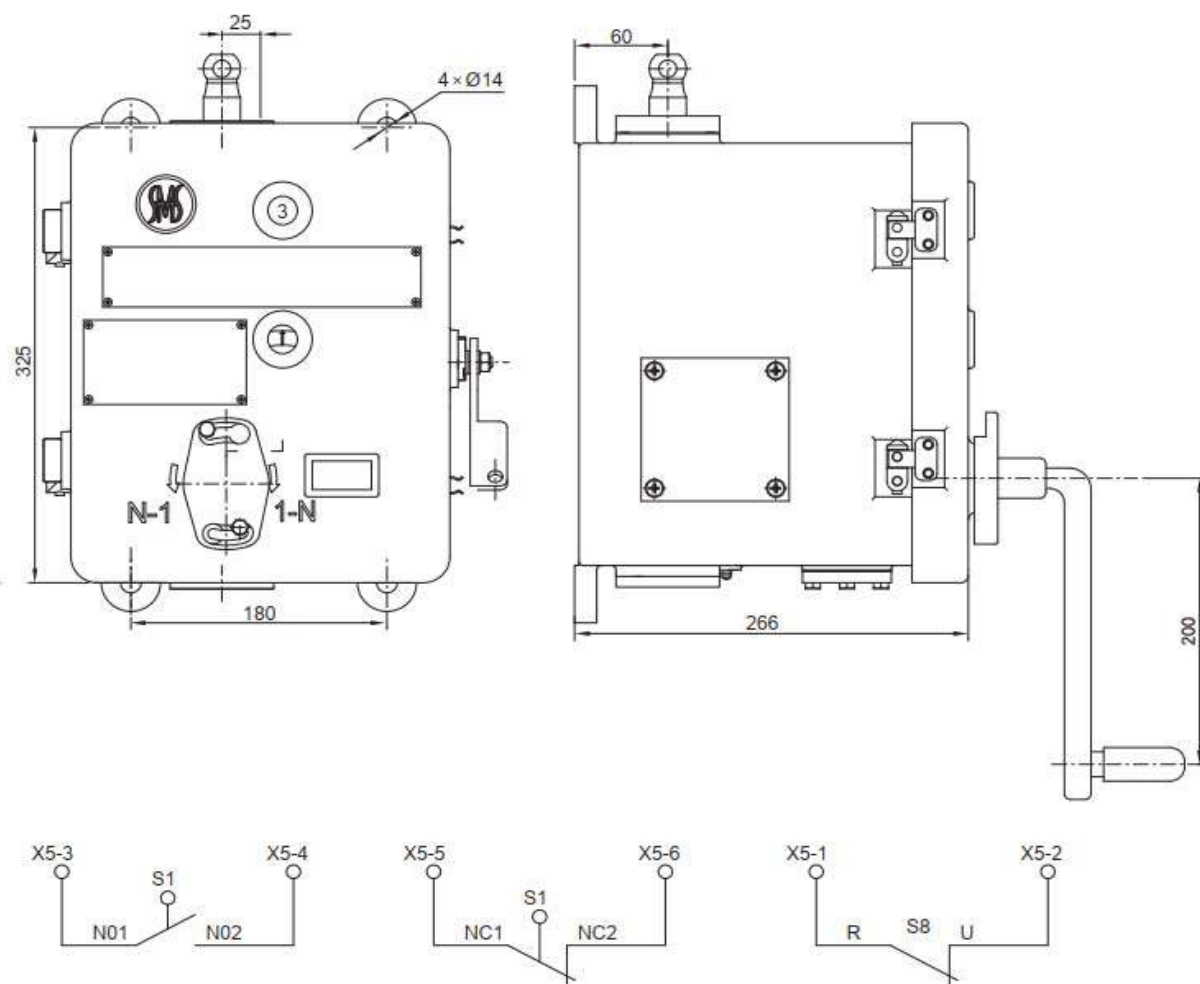
unit : mm

Appendix 28 Bevel gearbox dimensional drawing



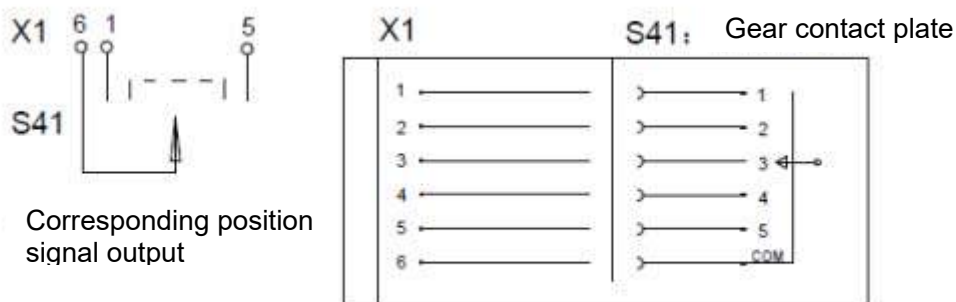
unit : mm

Appendix 29 Dimensional drawing of the manual



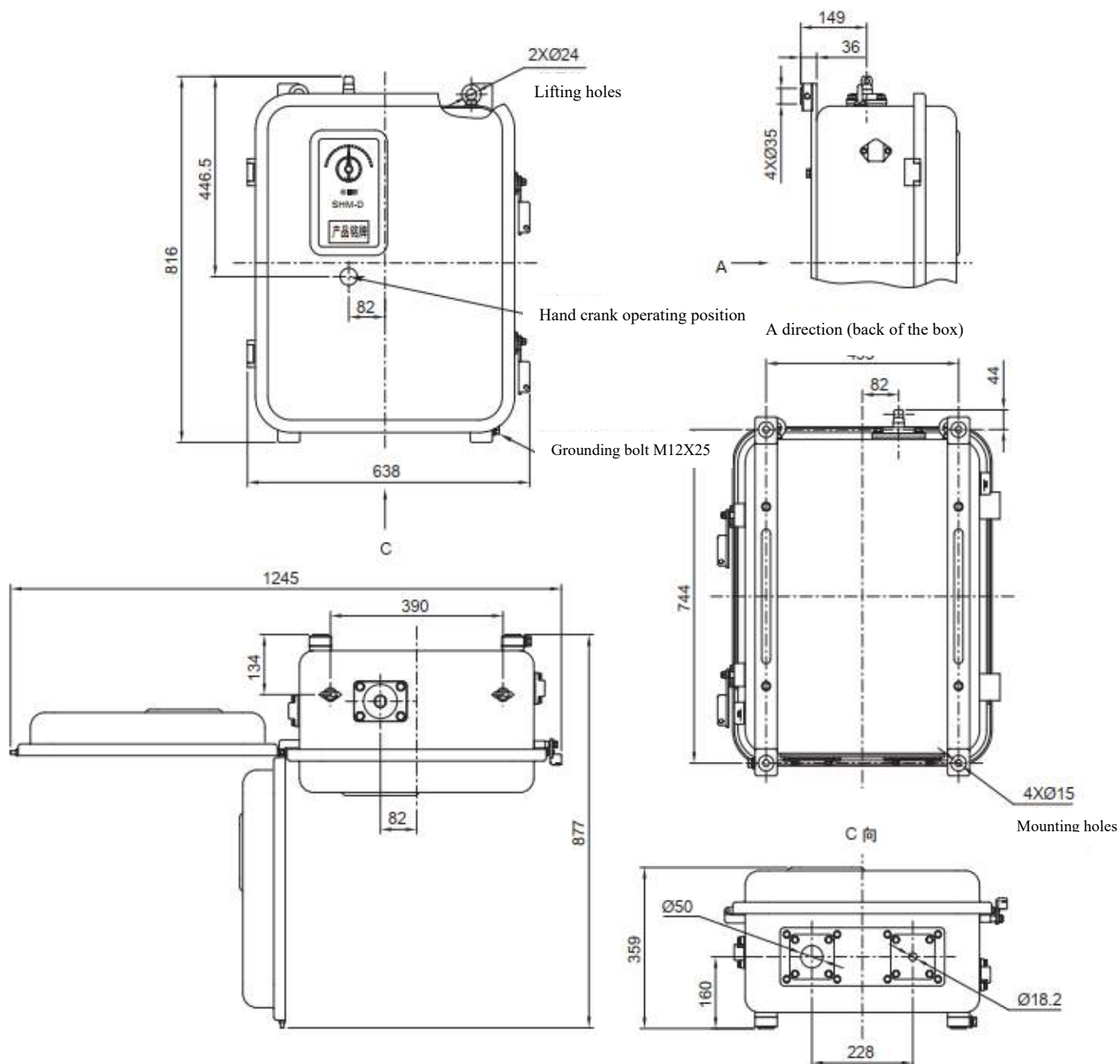
S1-NO1, S1-NO2 are signals that the mechanism is in motion. S1-NC1, S1-NC2 are signals that the mechanism is in position.

This hand crank mechanism leads S8-R, S8-U to terminals X5-1, X5-2 disconnected, if the operating handle is pulled out, then X5-1, X5-2 closed, the user should use this signal node for the handle mechanism and transformer circuit breaker blocking.

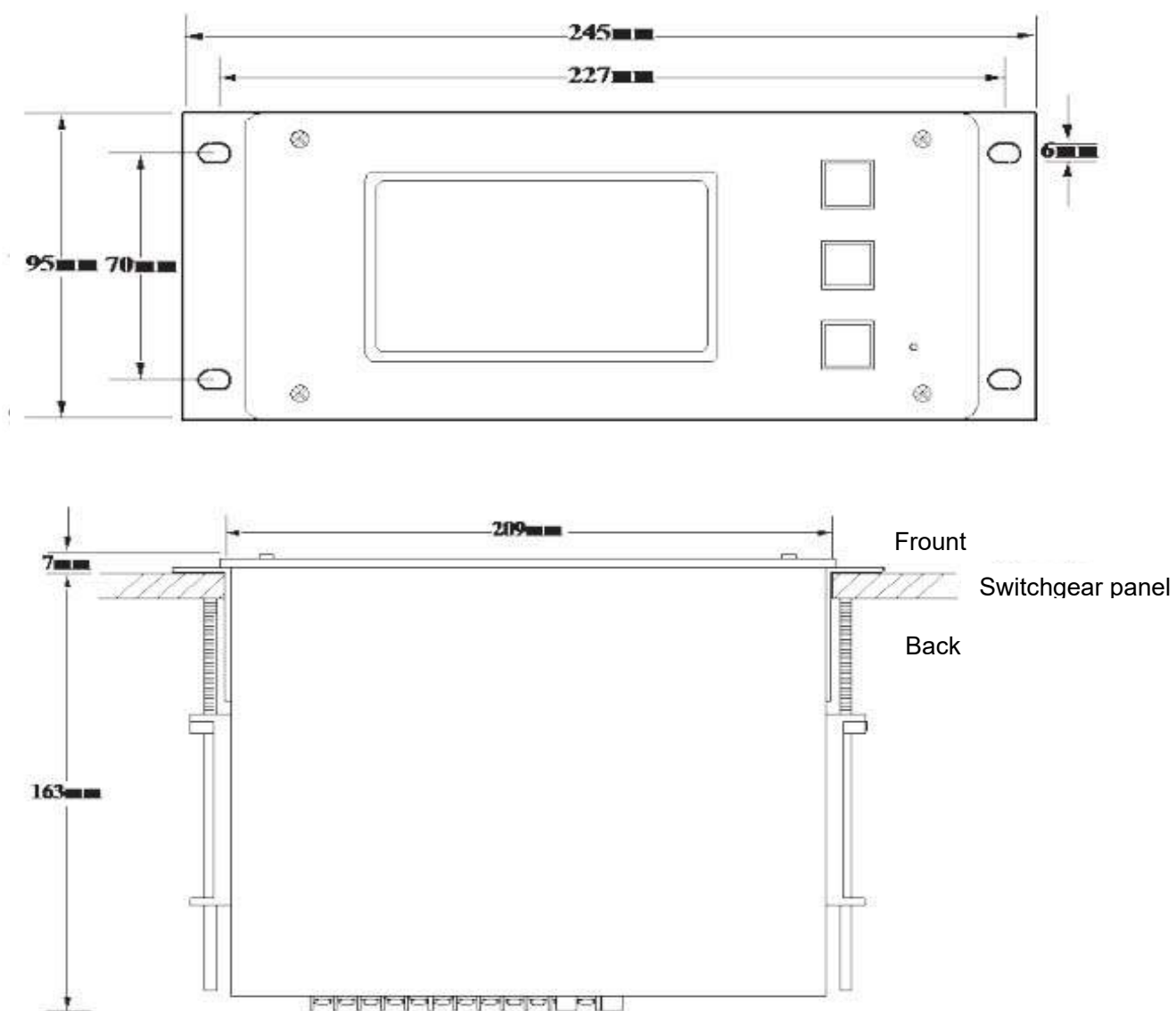


unit : mm

Appendix 30 SHM-D Electric motorised mechanism external mounting dimensions



Appendix 31 SHM-K Remote controller installation dimensions



**Material damage**

Damage to property and the environment.

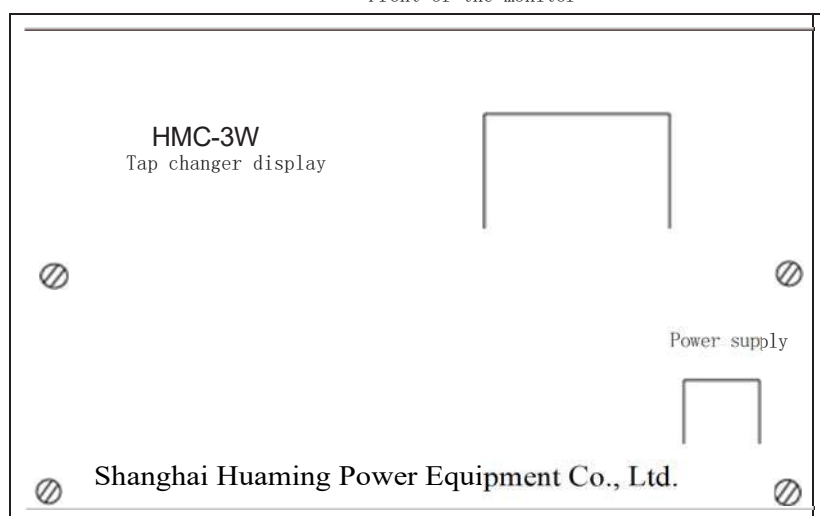
The mounting can be done with screws on the front.

Can also be installed by using the pin locking on both sides as shown

above Hole size: 210mm x 96mm (L x W)

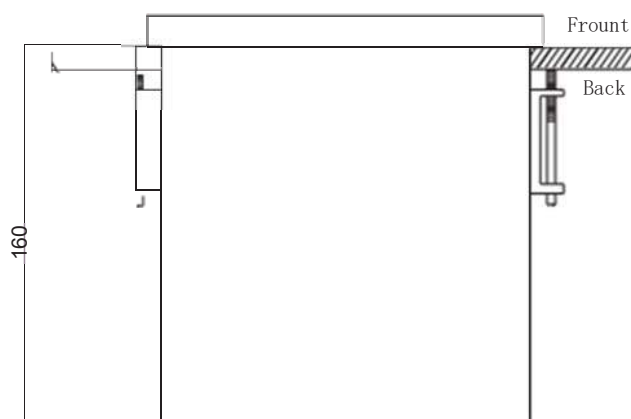
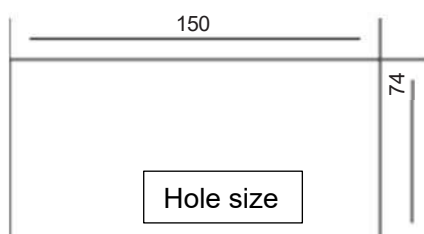
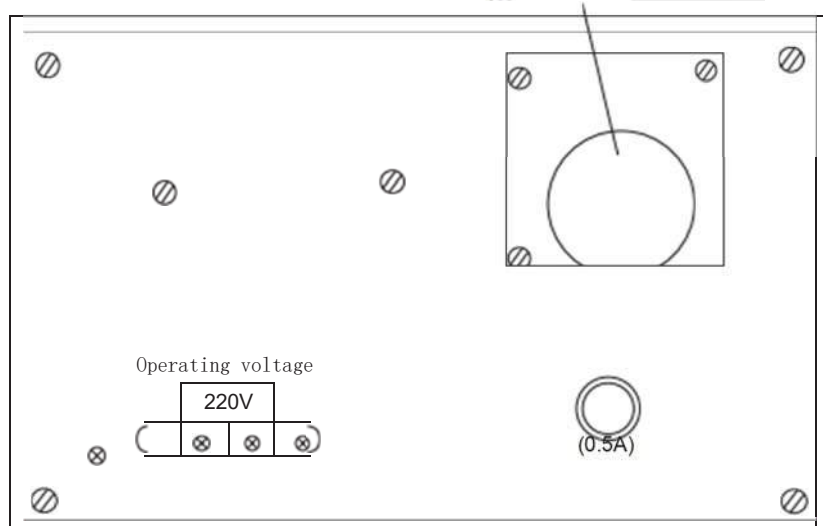
Appendix 32 HMC-3W Display form factor diagram

Front of the monitor



Back of monitor

Cable sockets



Unit:mm



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